

Dual Node Module Service

LC13-610

LCN Service - 1

***Dual Node Module
Service***

**LC13-610
Release 684
CE Compliant
March 2013**

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About This Publication

This publication provides instructions for use by system service personnel to service any Dual Node Module. It will help you to determine how to perform service that is required for Dual Node Modules, identify spare parts, and provide disassembly/assembly instructions for replacing the required part.

The manual is used to support the *Universal Station Service*, *Universal Station^X Service*, and *History Module Service* manuals.

The publication supports TPS software release 684, or earlier software, and CE Compliant hardware.

Any equipment designated as "CE Compliant" complies with the European Union EMC and Health and Safety Directives. All equipment shipping into European Union countries after January 1, 1996 requires this type of compliance—denoted by the "CE Mark."

This publication supports **TotalPlant** Solution (TPS) system network Release 684, or a later software release. TPS is the evolution of TDC 3000^X.

ATTENTION

With TPN R684, Enhanced Universal Control Network (EUCN) is introduced. In EUCN, the traditional coaxial cable system is no longer utilized and is replaced with a Fault Tolerant Ethernet (FTE) based communication network.

Throughout the document, unless specifically mentioned, the terms HPM, NIM, and UCN are interchangeably used with the terms EHPM, ENIM, and EUCN respectively.

Standard Symbols

Scope

The following defines standard symbols used in this publication.

ATTENTION

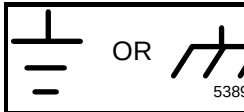
Notes inform the reader about information that is required, but not immediately evident.

CAUTION

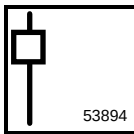
Cautions tell the user that damage may occur to equipment if proper care is not exercised.

WARNING

Warnings tell the reader that potential personal harm or serious economic loss may happen if instructions are not followed.



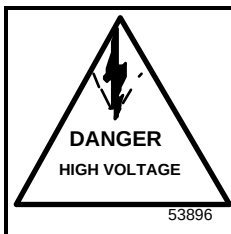
Ground connection to building safety ground



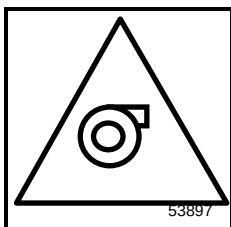
Ground stake for building safety ground



Electrical Shock Hazard—can be lethal



Electrical Shock Hazard—can be lethal



Rotating Fan—can cause personal injury

Table of Contents

1 INTRODUCTION

- 1.1 Overview
- 1.2 Related Publications

2 MODULE DESCRIPTION

- 2.1 General Description
- 2.2 Basic Module Configurations
- 2.3 Specific Node Configurations
- 2.4 Front Panel
- 2.5 Rear Panel Assembly
- 2.6 Logic and Chassis Ground
- 2.7 Field Adjustments
- 2.8 LCN Network Grounding
- 2.9 Precision Clock Pinning (Power Supply Replacement)
- 2.10 NIM Modem Board Pinning
- 2.11 EPNI2 I/O Board Pinning
- 2.12 Power Supply Loading in XLCNE2 Host Node

3 TEST/TROUBLESHOOTING

- 3.1 Tests
- 3.2 Test Procedures
 - 3.2.1 Power-On Test
 - 3.2.2 Reset Test
 - 3.2.3 Quality Logic Test
- 3.3 Troubleshooting
 - 3.3.1 Power Supply/Fan
 - 3.3.2 Controller Boards
 - 3.3.3 PLCI Board
 - 3.3.4 Processor Board (K2LCN)
 - 3.3.5 EPDG Board
 - 3.3.6 EPNI Board
 - 3.3.7 EPNI2 Board
 - 3.3.8 Fiber Optic Extender Boards
 - 3.3.9 RS-485 to LCN Converter Boards (KLCNA, KLCNB)
 - 3.3.10 NGI Board

4 DISASSEMBLY/ASSEMBLY

- 4.1 Disassembly
- 4.2 Assembly

5 SPARE PARTS

- 5.1 Introduction

Table of Contents

6 STARTUP

- 6.1 Visual Checks
- 6.2 Initialize Node

APPENDIX A

Alphanumeric Displays

- A.1 Recommended Actions for Specific Code Occurrences

APPENDIX B

Replacement of K2LCN with K4LCN Board

- B.1 Introduction
- B.2 Prerequisite
- B.3 Node Applicability
- B.4 Memory Size
- B.5 Redundant Module Replacements
- B.6 Disassembly
- B.7 Reassembly
- B.8 Test
- B.9 Startup

INDEX

INTRODUCTION

Section 1

1.1 OVERVIEW

This publication provides detailed instructions for maintenance, test, troubleshooting and repair of the Dual Node Module (part of the TPS system) that is illustrated in Figure 1-1. The troubleshooting, disassembly, and assembly procedures are effective down to the optimum replaceable-unit (ORU) level. An ORU parts list is included and is keyed to a module exploded view that is also used with the disassembly and assembly procedures.

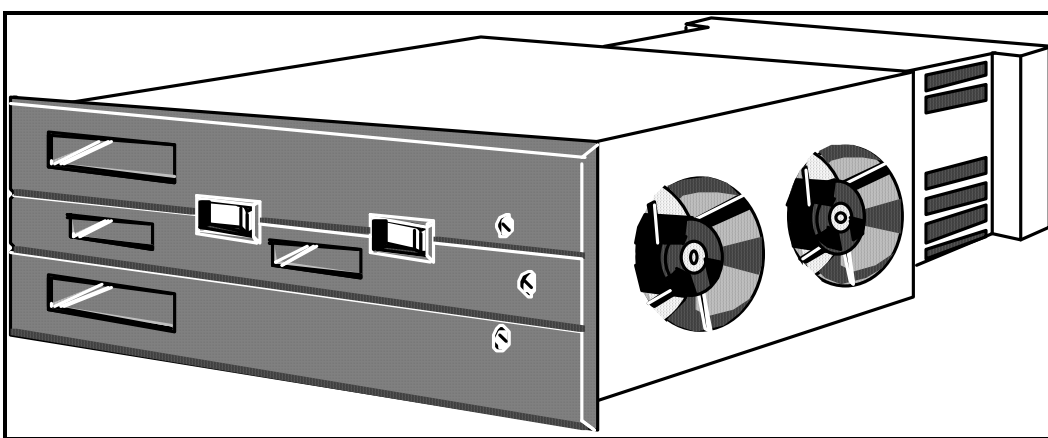


Figure 1-1 — Dual Node Module

4121

1.2 RELATED PUBLICATIONS

The following publications apply to the TPS system and must be referred to as required.

Title	Binder
LCN System Installation	LCN Installation
LCN System Checkout	LCN Installation
Universal Station Service	LCN Service - 1
Universal Station (Ergonomic) Service	LCN Service - 1
Universal Station ^X Service	Universal Station ^X
Universal Station ^X (Ergonomic) Service	Universal Station ^X
History Module Service	LCN Service - 2
Five/Ten-Slot Module Service	LCN Service - 2
Maintenance Test Operations	LCN Service - 1
System Maintenance Guide	LCN Service - 1
Application Module ^X Service	Application Module ^X

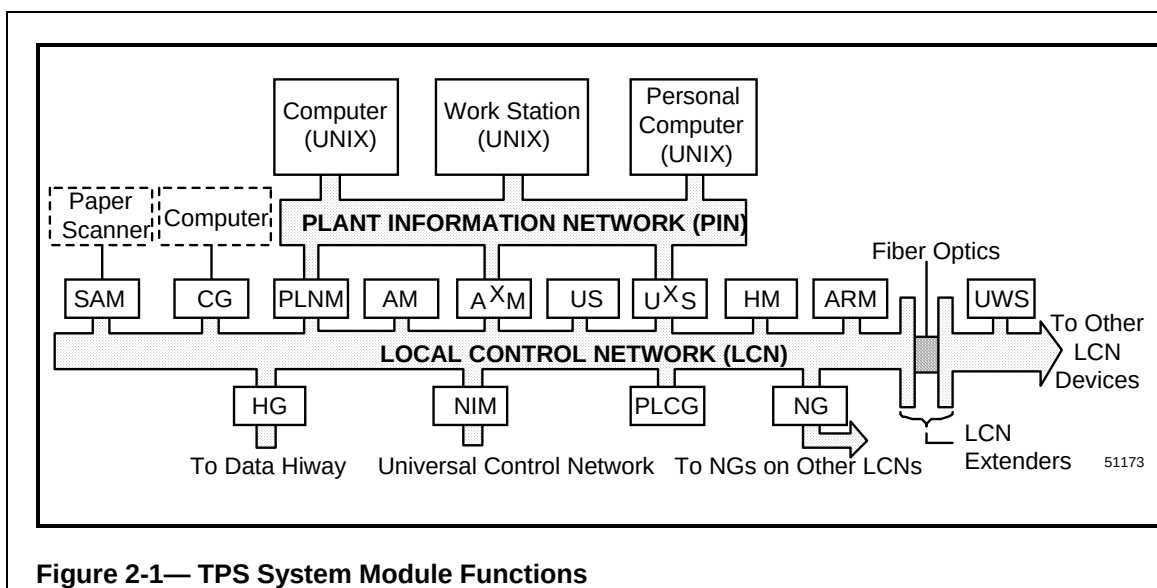
MODULE DESCRIPTION

Section 2

2.1 GENERAL DESCRIPTION

The Dual Node Module is a chassis containing one or two nodes. Each node performs an independent function and occupies a specific address on the Local Control Network (LCN). Figure 2-1 illustrates the module functions within a typical TPS system. The types of nodes available in the Dual Node Module include Application Modules, Network Interface Modules, History Modules, Universal Stations, Hiway Gateways, and other types of gateways.

The Dual Node Module can accommodate one three-board node and one two-board node. Each node has its own unique power supply—the Dual Node Power Supply (DNPWR). The Dual Node Module also includes a front panel, chassis, and a cooling fan assembly. The chassis consists of a card cage and backpanel assembly. The component cards (boards) mount in the front of the card cage assembly, and the associated input/output (I/O) boards mount in the rear, similar to the method used in Five- and Ten-Slot Modules.



The K2LCN or the K4LCN board is a common element of nodes implemented in the Dual Node Module. It combines the functions of kernel (processor), memory, and LCN (Local Control Network) interface in a single board. The K2LCN or K4LCN board is used in conjunction with KLCNA and KLCNB (non-CE Compliant), or CLCNA and CLCNB (CE Compliant) boards. The boards provide the physical interface to the LCN's redundant cables (the "A" and "B" cables) which interconnect the equipment in a TPS system control room. The CLCNA and CLCNB boards are installed in the I/O card cage at the rear of the module as illustrated in Table 2-1. One of each is required, regardless of whether one or both nodes are implemented.

2.2 BASIC MODULE CONFIGURATIONS

A Dual Node Module can accommodate one or two nodes. In a Dual Node Module, there is only one connection to the LCN A coaxial cable and only one connection to the LCN B cable. The connections are made on the KLCNA and KLCNB, or CLCNA and CLCNB boards, respectively, by using BNC T-connectors. The LCN signals are communicated between nodes within a Dual Node Module by RS-485 signals on the backpanel. RS-485 is an electrical signaling standard. The KLCNA and KLCNB, or CLCNA and CLCNB boards, provide interface conversion between the RS-485 backpanel signals and the LCN network.

Table 2-1 illustrates the basic (minimum) module configurations for single- and double-node modules. The Application Module (AM) uses the basic configuration. Other module types require additional boards, as covered in subsection 2.3. Figure 2-2 is a rear view of a Dual Node Module showing the LCN A and LCN B coaxial cable interconnects and the RS-485 terminator assemblies that are installed on the backpanel at the top and bottom ends of the RS-485 signal conductors. Four terminator assemblies are required—two at the top and two at the bottom of the backpanel.

CAUTION

Slot 1 (the lowest) in the input/output (I/O) card cage for each node is normally reserved for CLCNA and CLCNB boards. Do not insert any other type of board into these slots unless specifically authorized by Honeywell documentation, or damage may result. This is particularly true for CSR boards—DO NOT insert a CSR board in I/O Slot 1.

As illustrated in Table 2-1, the CLCNA and CLCNB boards are installed in different nodes in a Dual Node Module. The CLCNA is installed in the upper node, and the CLCNB is installed in the lower node. Each is installed in Slot 1 (the lowest slot) in the I/O card cage of its respective node. The CLCNA and CLCNB boards each receive power from both power supplies in the Dual Node Module. Therefore, if one power supply is not present, or is turned off to perform troubleshooting or maintenance on one node, both the CLCNA and CLCNB boards will receive power and the node which does have power will continue to have access to both the LCN A and LCN B cables in the LCN network. The CLCNA and CLCNB are the only boards in the I/O card cage which are powered from both power supplies. The Fan Assembly is also powered from both power supplies. It will operate if either supply is on (or if both are on). The CLCNA and CLCNB boards are designed to be able to be plugged into and unplugged from the system while power is on, allowing them to be changed without removing power from a node. Do not attempt to do this with any other board type.

Table 2-1— Basic Module Configurations

Table 1-1 Basic Module Configurations

Dual Node Module with One Node Installed			
Upper Node	Slot	Front	Rear
	2		
	1		CLCNA

Power Supply	Left	Right	Fan Module
		DNPWR	

Lower Node	Slot	Front	Rear
	3		
	2		
	1	K2LCN OR K4LCN	CLCNB

Dual Node Module with Two Nodes Installed			
Upper Node	Slot	Front	Rear
	2		
	1	K2LCN OR K4LCN	CLCNA

Power Supply	Left	Right	Fan Module
	DNPWR	DNPWR	

Lower Node	Slot	Front	Rear
	3		
	2		
	1	K2LCN OR K4LCN	CLCNB

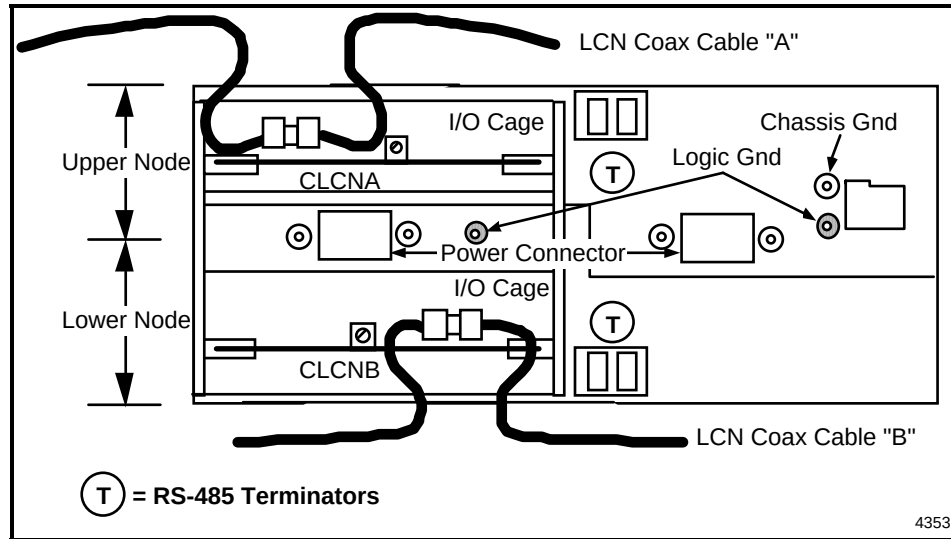


Figure 2-2— Rear View of Dual Node Module

2.3 SPECIFIC NODE CONFIGURATIONS

Tables 2-2 and 2-3 illustrate the board configurations for the different types of Dual Node Module nodes. Table 2-2 illustrates configurations available in the upper, two-slot node, and Table 2-3 illustrates the configurations available in lower, three-slot node.

Table 2-2— Two-Slot Node Configurations

Application Module (68020)			Application Module (68020) CE Compliant		
Slot	Front	Rear	Slot	Front	Rear
2		(Note 1)	2		(Note 1)
1	K2LCN	KLCNA	1	K2LCN	CLCNA

Computer Gateway (68020)			Computer Gateway (68020) CE Compliant		
Slot	Front	Rear	Slot	Front	Rear
2	CLI	CLI I/O	2	CLI	CLI I/O
1	K2LCN	KLCNA	1	K2LCN	CLCNA

History Module (68020)			History Module (68020) CE Compliant		
Slot	Front	Rear	Slot	Front	Rear
2	SPC	SPC I/O	2	SPC	SPC -2
1	K2LCN	KLCNA	1	K2LCN	CLCNA

Hiway Gateway (68020) (See Note 6)			Hiway Gateway (68020) CE Compliant (See Note 6)		
Slot	Front	Rear	Slot	Front	Rear
2	DHI	DHI I/O	2	DHI	DHI
1	K2LCN	KLCNA	1	K2LCN	CLCNA

(Continued)

Note 1 Can be used for FOCT, FOCT, LCNE, or XLCNE2.

Note 2 Redundant units require Relay Switch Board 51304421-100.

Note 3 Cartridge Disks are optional (via EPDG).

Note 4 The EPNI board is a later version of the PNI board.

Note 5 This Network Gateway configuration can never be expanded to a redundant cable configuration.

Note 6 HG permitted if the other node is a redundant HG for another Data Hiway.

Table 2-2— Two-Slot Node Configurations. Continued

Network Gateway (NG) (68020) <i>(Note 5)</i>			Network Gateway (NG) (68020) CE Compliant <i>(Note 5)</i>		
Slot	Front	Rear	Slot	Front	Rear
2	NGI	NGIO	2	NGI	NGI
1	K2LCN	KLCNA	1	K2LCN	CLCNA

Network Interface Module (68020) <i>(Note 4)</i>			Network Interface Module (68020) CE Compliant <i>(Note 4)</i>		
Slot	Front	Rear	Slot	Front	Rear
2	EPNI	NIM MODEM	2	EPNI	NIM
1	K2LCN	KLCNA	1	K2LCN	CLCNA

Enhanced Network Interface Module (68020) <i>(Note 4a, Note 4b and Note 4c)</i>			Enhanced Network Interface Module (68020) CE Compliant <i>(Note 4a, Note 4b and Note 4c)</i>		
Slot	Front	Rear	Slot	Front	Rear
2	EPNI2	EPNI2 I/O	2	EPNI2	EPNI2 I/O
1	K2LCN	KLCNA	1	K2LCN	CLCNA

PLC Gateway (PLCG) (68020) <i>(Note 2)</i>			PLC Gateway (PLCG) (68020) CE Compliant <i>(Note 2)</i>		
Slot	Front	Rear	Slot	Front	Rear
2	PLCI	PLCI I/O	2	PLCI	PLCI
1	K2LCN	KLCNA	1	K2LCN	CLCNA

Universal Station (68020) <i>(Note 3)</i>			Universal Station (68020) CE Compliant <i>(Note 3)</i>		
Slot	Front	Rear	Slot	Front	Rear
2	EPDG	EPDGP I/O	2	EPDG	EPDGC I/O
1	K2LCN	KLCNA	1	K2LCN	CLCNA

Note 1 Can be used for FOCT, FOCT, LCNE, or XLCNE2.

Note 2 Redundant units require Relay Switch Board 51304421-100.

Note 3 Cartridge Disks are optional (via EPDG).

Note 4 The EPNI board is a later version of the PNI board.

Note 4a The EPNI2 board is a later version of the EPNI board.

Note 4b The minimum supported LCN board for ENIM is K2LCN-3.

Note 4c ENIM does not support LCN boards of HMPU and HPK2 types.

Note 5 This Network Gateway configuration can never be expanded to a redundant cable configuration.

Note 6 HG permitted if the other node is a redundant HG for another Data Hiway

Table 2-2— Two-Slot Node Configurations. Continued

Application Module (68040)			Application Module (68040) CE Compliant		
Slot	Front	Rear	Slot	Front	Rear
2		(Note 1)	2		(Note 1)
1	K4LCN	KLCNA	1	K4LCN	CLCNA

Computer Gateway (68040)			Computer Gateway (68040) CE Compliant		
Slot	Front	Rear	Slot	Front	Rear
2	CLI	CLI I/O	2	CLI	CLI I/O
1	K4LCN	KLCNA	1	K4LCN	CLCNA

History Module (68040)			History Module (68040) CE Compliant		
Slot	Front	Rear	Slot	Front	Rear
2	SPC	SPC I/O	2	SPC	SPC -2
1	K4LCN	KLCNA	1	K4LCN	CLCNA

Hiway Gateway (68040) (See Note 6)			Hiway Gateway (68040) CE Compliant (See Note 6)		
Slot	Front	Rear	Slot	Front	Rear
2	DHI	DHI I/O	2	DHI	DHI
1	K4LCN	KLCNA	1	K4LCN	CLCNA

Network Gateway (NG) (68040) (Note 5)			Network Gateway (NG) (68040) CE Compliant (Note 5)		
Slot	Front	Rear	Slot	Front	Rear
2	NGI	NGIO	2	NGI	NGI
1	K4LCN	KLCNA	1	K4LCN	CLCNA

(Continued)

Note 1 Can be used for FOCT, FOCT, LCNE, or XLCNE2.

Note 2 Redundant units require Relay Switch Board 51304421-100.

Note 3 Cartridge Disks are optional (via EPDG).

Note 4 The EPNI board is a later version of the PNI board.

Note 5 This Network Gateway configuration can never be expanded to a redundant cable configuration.

Note 6 HG permitted if the other node is a redundant HG for another Data Hiway.

Table 2-2— Two-Slot Node Configurations, Continued

Network Interface Module (68040) <i>(Note 4)</i>			Network Interface Module (68040) CE Compliant <i>(Note 4)</i>		
Slot	Front	Rear	Slot	Front	Rear
2	EPNI	NIM MODEM	2	EPNI	NIM
1	K4LCN	KLCNA	1	K4LCN	CLCNA

Enhanced Network Interface Module (68040) <i>(Note 4a)</i>			Enhanced Network Interface Module (68040) CE Compliant <i>(Note 4a)</i>		
Slot	Front	Rear	Slot	Front	Rear
2	EPNI2	EPNI2 I/O	2	EPNI2	EPNI2 I/O
1	K4LCN	KLCNA	1	K4LCN	CLCNA

PLC Gateway (PLCG) (68040) <i>(Note 2)</i>			PLC Gateway (PLCG) (68040) CE Compliant <i>(Note 2)</i>		
Slot	Front	Rear	Slot	Front	Rear
2	PLCI	PLCI I/O	2	PLCI	PLCI
1	K4LCN	KLCNA	1	K4LCN	CLCNA

Universal Station (68040) <i>(Note 3)</i>			Universal Station (68040) CE Compliant <i>(Note 3)</i>		
Slot	Front	Rear	Slot	Front	Rear
2	EPDG	EPDGP I/O	2	EPDG	EPDGC I/O
1	K4LCN	KLCNA	1	K4LCN	CLCNA

Note 1 Can be used for FOCT, FOCR, LCNE, or XLCNE2.

Note 2 Redundant units require Relay Switch Board 51304421-100.

Note 3 Cartridge Disks are optional (via EPDG).

Note 4 The EPNI board is a later version of the PNI board.

Note 4a The EPNI2 board is a later version of the EPNI board.

Note 5 This Network Gateway configuration can never be expanded to a redundant cable configuration.

Note 6 HG permitted if the other node is a redundant HG for another Data Hiway.

Table 2-3— Three-Slot Node Configurations

Application Module (68020)			Application Module CE Compliant (68020)		
Slot	Front	Rear	Slot	Front	Rear
3		(Note 1)	3		(Note 1)
2		(Note 1)	2		(Note 1)
1	K2LCN	KLCNB	1	K2LCN	CLCNB

Computer Gateway (68020)			Computer Gateway CE Compliant (68020)		
Slot	Front	Rear	Slot	Front	Rear
3	CLI	CLI I/O	3	CLI	CLI
2		(Note 1)	2		(Note 1)
1	K2LCN	KLCNB	1	K2LCN	CLCNB

History Module (68020)			History Module CE Compliant (68020)		
Slot	Front	Rear	Slot	Front	Rear
3	SPC	SPC I/O	3	SPC	SPC
2		(Note 1)	2		(Note 1)
1	K2LCN	KLCNB	1	K2LCN	CLCNB

Hiway Gateway (68020) (Note 2)			Hiway Gateway (68020) CE Compliant (Note 2)		
Slot	Front	Rear	Slot	Front	Rear
3	DHI	DHI I/O	3	DHI	DHI
2		(Note 1)	2		(Note 1)
1	K2LCN	KLCNB	1	K2LCN	CLCNB

Network Gateway (NG) (68020) (Note 8)			Network Gateway (NG) (68040) CE Compliant (Note 8)		
Slot	Front	Rear	Slot	Front	Rear
3	NGI	NGIO	3	NGI	NGI or NGFOM
2	NGI	NGIO	2	NGI	NGI or NGFOM
1	K2LCN	KLCNA	1	K2LCN	CLCNB

(Continued)

Note 1 Can be used for FOCT, FOCR, LCNE, or XLCNE2.

Note 2 Hiway Gateway can only be combined with another redundant Hiway Gateway node connected to another Data Hiway. Logic and chassis ground must be isolated on a Hiway Gateway Dual Node Module. (Both halves of a Dual Node Module have a common ground system).

Note 8 The NGI and NGIO in Slot 3 are optional. They are present for a redundant cable NG.

Table 2-3— Three-Slot Node Configurations, Continued

Network Gateway (NG) (68020) (Note 8)			Network Gateway (NG) (68020) CE Compliant (Note 8)		
Slot	Front	Rear	Slot	Front	Rear
3	NGI	NGIO	3	NGI	NGI or NGFOM
2	NGI	NGIO	2	NGI	NGI or NGFOM
1	K2LCN	KLCNA	1	K2LCN	CLCNB

Network Interface Module (68020) (Note 7)			Network Interface Module (68020) CE Compliant (Note 7)		
Slot	Front	Rear	Slot	Front	Rear
3	EPNI	NIM MODEM	3	EPNI	NIM
2		(Note 1)	2		(Note 1)
1	K2LCN	KLCNB	1	K2LCN	CLCNB

Plant Network Module (PLNM) (68020)			Plant Network Module (PLNM) CE Compliant (68020)		
Slot	Front	Rear	Slot	Front	Rear
3	(Note 6)	(Note 1)	3	(Note 6)	(Note 1)
2	CNI	CNI I/O	2	CNI	CNI
1	K2LCN	KLCNB	1	K2LCN	CLCNB

PLC Gateway (PLCG) (68020) (Note 3)			PLC Gateway (PLCG) (68020) CE Compliant (Note 3)		
Slot	Front	Rear	Slot	Front	Rear
3	PLCI	PLCI I/O	3	PLCI	PLCI
2		(Note 1)	2		(Note 1)
1	K2LCN	KLCNB	1	K2LCN	CLCNB

(Continued)

Note 3 Redundant units require Relay Switch Board 51304421-100.

Note 6 Slot not available due to height of CNI board.

Note 7 The EPNI board is a later version of the PNI board.

Note 8 The NGI and NGIO in Slot 3 are optional. They are present for a redundant cable NG.

Table 2-3— Three-Slot Node Configurations, Continued

Universal Station (68020) (Note 4)			Universal Station (68020) CE Compliant (Note 4)		
Slot	Front	Rear	Slot	Front	Rear
3	EPDG	EPDGP I/O	3	EPDG	EPDGC
2		(Note 1)	2		(Note 1)
1	K2LCN	KLCNB	1	K2LCN	CLCNB

Universal Station with Floppy Drives (68020) (Note 5)			Universal Station with Floppy Drives (68020) CE Compliant (Note 5)		
Slot	Front	Rear	Slot	Front	Rear
3	EPDG	EPDGP I/O	3	EPDG	EPDGC
2	FDC	FDC I/O	2	FDC	FDC I/O
1	K2LCN	KLCNB	1	K2LCN	CLCNB

Universal Station with Trend Pen (68020) (Note 4, Note 5)			Universal Station with Trend Pen (68020) CE Compliant (Note 4, Note 5)		
Slot	Front	Rear	Slot	Front	Rear
3	EPDG	EPDGP I/O	3	EPDG	EPDGC
2	SIO	SIO I/O	2	SIO	SIO I/O
1	K2LCN	KLCNB	1	K2LCN	CLCNB

Note 1 Can be used for FOCT, FOCR, LCNE, or XLCNE2.

Note 4 Cartridge disks are optional (via EPDG)

Note 5 This Network Gateway configuration can never be expanded to a redundant cable configuration.

Table 2-3— Three-Slot Node Configurations, Continued

Application Module (68040)			Application Module CE Compliant (68040)		
Slot	Front	Rear	Slot	Front	Rear
3		(Note 1)	3		(Note 1)
2		(Note 1)	2		(Note 1)
1	K4LCN	KLCNB	1	K4LCN	CLCNB

Computer Gateway (68040)			Computer Gateway CE Compliant (68040)		
Slot	Front	Rear	Slot	Front	Rear
3	CLI	CLI I/O	3	CLI	CLI
2		(Note 1)	2		(Note 1)
1	K4LCN	KLCNB	1	K4LCN	CLCNB

History Module (68040)			History Module CE Compliant (68040)		
Slot	Front	Rear	Slot	Front	Rear
3	SPC	SPC I/O	3	SPC	SPC
2		(Note 1)	2		(Note 1)
1	K4LCN	KLCNB	1	K4LCN	CLCNB

Hiway Gateway (68040) (Note 2)			Hiway Gateway (68040) CE Compliant (Note 2)		
Slot	Front	Rear	Slot	Front	Rear
3	DHI	DHI I/O	3	DHI	DHI
2		(Note 1)	2		(Note 1)
1	K4LCN	KLCNB	1	K4LCN	CLCNB

Network Gateway (NG) (68040) (Note 8)			Network Gateway (NG) (68040) CE Compliant (Note 8)		
Slot	Front	Rear	Slot	Front	Rear
3	NGI	NGIO	3	NGI	NGI or NGFOM
2	NGI	NGIO	2	NGI	NGI or NGFOM
1	K4LCN	KLCNA	1	K4LCN	CLCNB

(Continued)

Note 1 Can be used for FOCT, FOCR, LCNE, or XLCNE2.

Note 2 Hiway Gateway can only be combined with another redundant Hiway Gateway node connected to another Data Hiway. Logic and chassis ground must be isolated on a Hiway Gateway Dual Node Module. (Both halves of a Dual Node Module have a common ground system).

Note 8 The NGI and NGIO in Slot 3 are optional. They are present for a redundant cable NG.

Table 2-3— Three-Slot Node Configurations, Continued

Network Gateway (NG) (68040) (Note 8)			Network Gateway (NG) (68040) CE Compliant (Note 8)		
Slot	Front	Rear	Slot	Front	Rear
3	NGI	NGIO	3	NGI	NGI or NGFOM
2	NGI	NGIO	2	NGI	NGI or NGFOM
1	K4LCN	KLCNA	1	K4LCN	CLCNB

Network Interface Module (68040) (Note 7)			Network Interface Module (68040) CE Compliant (Note 7)		
Slot	Front	Rear	Slot	Front	Rear
3	EPNI	NIM MODEM	3	EPNI	NIM
2		(Note 1)	2		(Note 1)
1	K4LCN	KLCNB	1	K4LCN	CLCNB

Plant Network Module (PLNM) (68040)			Plant Network Module (PLNM) CE Compliant (68040)		
Slot	Front	Rear	Slot	Front	Rear
3	(Note 6)	(Note 1)	3	(Note 6)	(Note 1)
2	CNI	CNI I/O	2	CNI	CNI
1	K4LCN	KLCNB	1	K4LCN	CLCNB

PLC Gateway (PLCG) (68040) (Note 3)			PLC Gateway (PLCG) (68040) CE Compliant (Note 3)		
Slot	Front	Rear	Slot	Front	Rear
3	PLCI	PLCI I/O	3	PLCI	PLCI
2		(Note 1)	2		(Note 1)
1	K4LCN	KLCNB	1	K4LCN	CLCNB

(Continued)

Note 3 Redundant units require Relay Switch Board 51304421-100.

Note 6 Slot not available due to height of CNI board.

Note 7 The EPNI board is a later version of the PNI board.

Note 8 The NGI and NGIO in Slot 3 are optional. They are present for a redundant cable NG.

Table 2-3— Three-Slot Node Configurations, Continued

Universal Station (68040) (Note 4)			Universal Station (68040) CE Compliant (Note 4)		
Slot	Front	Rear	Slot	Front	Rear
3	EPDG	EPDGP I/O	3	EPDG	EPDGC
2		(Note 1)	2		(Note 1)
1	K4LCN	KLCNB	1	K4LCN	CLCNB

Universal Station with Floppy Drives (68040) (Note 5)			Universal Station with Floppy Drives (68040) CE Compliant (Note 5)		
Slot	Front	Rear	Slot	Front	Rear
3	EPDG	EPDGP I/O	3	EPDG	EPDGC
2	FDC	FDC I/O	2	FDC	FDC I/O
1	K4LCN	KLCNB	1	K4LCN	CLCNB

Universal Station with Trend Pen (68040) (Note 4, Note 5)			Universal Station with Trend Pen (68040) CE Compliant (Note 4, Note 5)		
Slot	Front	Rear	Slot	Front	Rear
3	EPDG	EPDGP I/O	3	EPDG	EPDGC
2	SIO	SIO I/O	2	SIO	SIO I/O
1	K4LCN	KLCNB	1	K4LCN	CLCNB

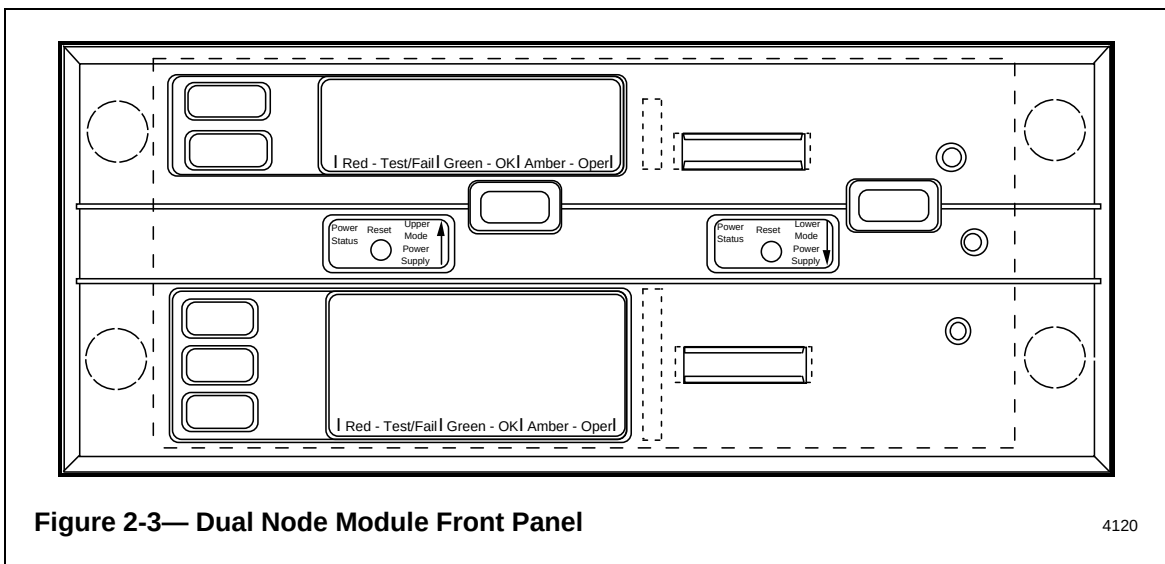
Note 1 Can be used for FOCT, FOCR, LCNE, or XLCNE2.

Note 4 Cartridge disks are optional (via EPDG)

Note 5 This Network Gateway configuration can never be expanded to a redundant cable configuration.

2.4 FRONT PANEL

The module front panel has the necessary windows and switch openings to accommodate two nodes in a single module. Figure 2-3 shows the front panel for the Dual Node Module. For each of the two power supplies, there are openings for the POWER switch and RESET switch and a window to view the light-emitting diode (LED) status indicators. Each power supply has a MARGIN pin jumper option which is a test/diagnostic aid and should be left in the center (N) position at all times. There is also a LED indicator at the right-center of the panel which indicates a fan module failure. For each of the two nodes, the front panel includes an access window to view the indicators that are mounted on the individual circuit boards. These indicators serve to monitor node performance and provide diagnostic information for fault isolation. The detailed functions and operation of the switches and indicators are discussed elsewhere in this manual.



2.5 REAR PANEL ASSEMBLY

The Rear Panel Assembly includes the backpanel Printed Wire Assembly (PWA) and the Input/Output (I/O) Card Cage Assembly. The power supplies, the fan assembly and the main circuit boards (such as the K2LCN or K4LCN) all plug into card edge connectors on the front side of the backpanel PWA. The I/O boards all plug into card edge connectors on the rear side of the backpanel PWA. As shown in Tables 2-2 and 2-3, the I/O boards are normally installed in the chassis in the slot corresponding in number to the applicable board installed in the front of the node. The CLCNA (KLCNA) and CLCNB (KLCNB) boards are installed as shown in Table 2-1. The various I/O boards support interface to a variety of external networks and devices such as the LCN, Data Hiway, keyboards, monitors, disk drives, and recorders.

Other connections available at the rear panel assembly are:

- **Ac Input**—The ac input cable for each power supply terminates in a receptacle which is attached to the backpanel support at the rear of the power supply. The power supply plugs into the receptacle when it is inserted into the chassis. The receptacles pass through, but do not connect to, the backpanel.
- **Remote Reset**—There is a 3-pin connector soldered to the backpanel for each node which permits the node to be reset by a remotely-located switch.
- **External Clock**—There is a 2-pin connector soldered to the backpanel for each node which allows use of an external precision clock source for the system clock.
- **RS-485 Connectors**—The RS-485 bus communicates the LCN A and B signals between the two nodes in a Dual Node Module. There are connectors at the upper and lower ends of this bus. They provide a place to install resistor terminator assemblies. Two terminator assemblies are installed at the top and two are installed at the bottom.

2.6 LOGIC AND CHASSIS GROUND

There are two threaded posts for logic ground on the Rear Panel Assembly – one behind each power supply. In most applications, logic ground is connected to chassis ground, and this connection is made by installing a jumper wire between the logic ground post and the rightmost mounting screw of the Ac power connector. This connection is made for both nodes in the module. Figure 2-7 and Figure 8 shows these logic-to-chassis ground jumpers installed in two modules. If isolated logic ground is required, the jumpers are not installed. Note that both nodes have a common logic ground through the RS-485 bus, and therefore a single node cannot be ground-isolated by itself. Both nodes must be either connected to chassis ground or isolated from chassis ground, but they cannot be ground-isolated with respect to each other. If the nodes are to be isolated from chassis ground, both jumpers must be removed.

There is also a threaded hole on the Rear Panel Assembly. This is a connection point for chassis ground. Figure 2-2 also shows the location of the logic ground and chassis ground points on the Rear Panel Assembly. These connection points are also physically marked for identification on the Rear Panel Assembly.

2.7 FIELD ADJUSTMENTS

There are no field adjustments for the Dual Node Module. However, the K2LCN and K4LCN boards must be pinned for the particular node address they occupy on the LCN. Figures 2-4 through 2-7 illustrate the pinning of the K2LCN and K4LCN boards. The EPDGC I/O board, if present, has pinning options to set the default background shade for the CRT if a palette has not been set in a schematic (Set Palette is a new command in Release 320). You can find additional information on the Set Palette command in the *Picture Editor Reference Manual*. The EPDGC board also has a configuration option that is set for either the Engineer's Keyboard or the Supervisor's Keyboard. (If both keyboards are installed, the EPDGC board is set up for the Supervisor's Keyboard and the Engineer's Keyboard is connected into the Supervisor's Keyboard.) Figure 2-8 illustrates the keyboard and CRT background options for the EPDGC board. Also, the power supply has a pinning option for internal or external clock. See subsection 2.9 and Figure 2-11 for additional information. The PLCI board has several pinning options. Refer to the *Programmable Logic Controller Gateway Planning, Installation, and Service* manual for details. Figure 2-12 in subsection 2.10 covers UCN address pinning for the NIM Modem board.

Figure 2-5 illustrates LCN node address selection on the K2LCN board, assembly 51402615-x00. The pinning block, TS1, is located at coordinate 7A on the board. The absence of a jumper is a binary “1.” Parity must be an odd number.

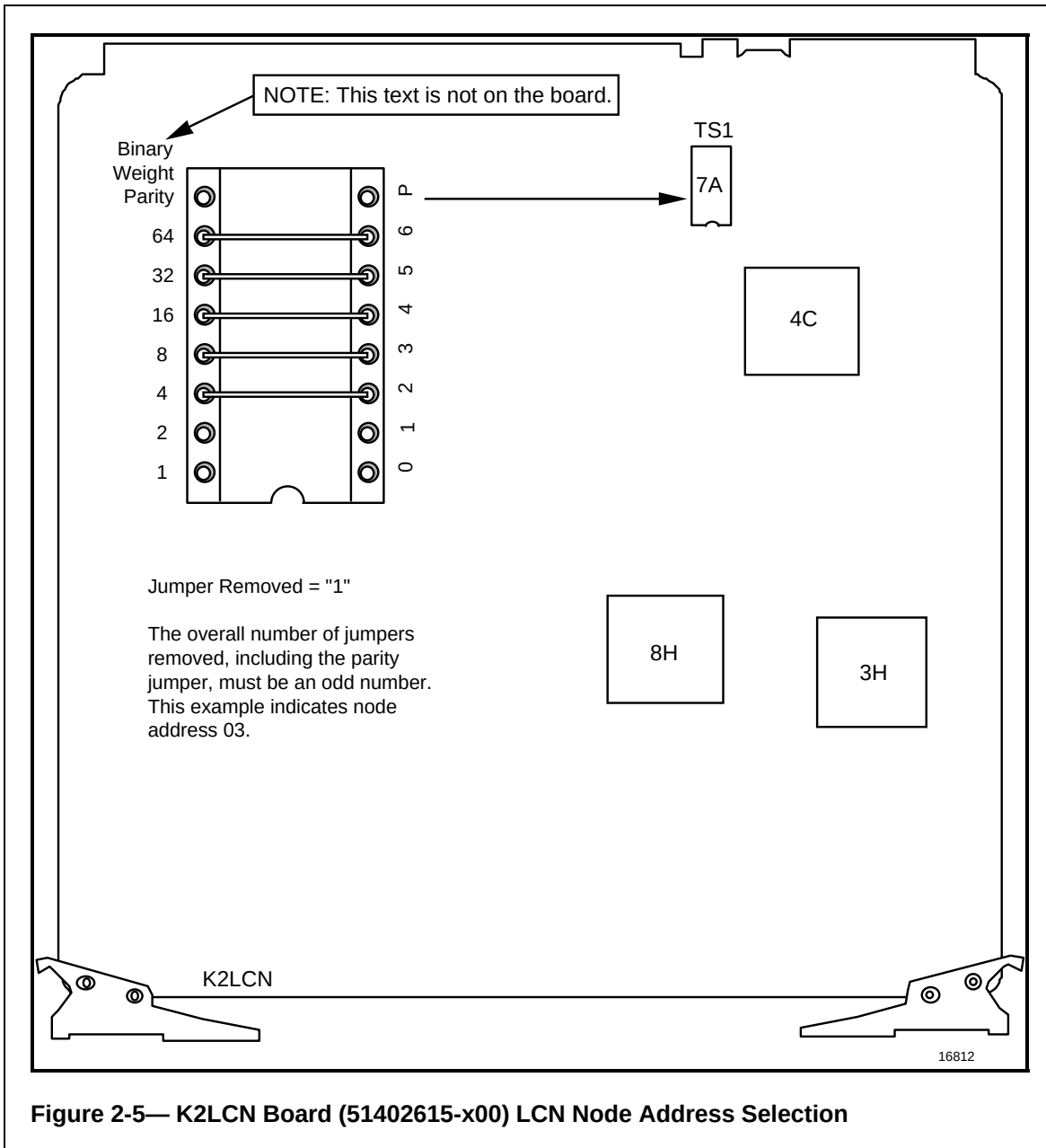


Figure 2-6 illustrates LCN node address selection on the K4LCN board, assembly 51401946-100. The pinning block, TS2, is located at coordinate 21E on the board. The absence of a jumper is a binary “1.” Parity must be an odd number.

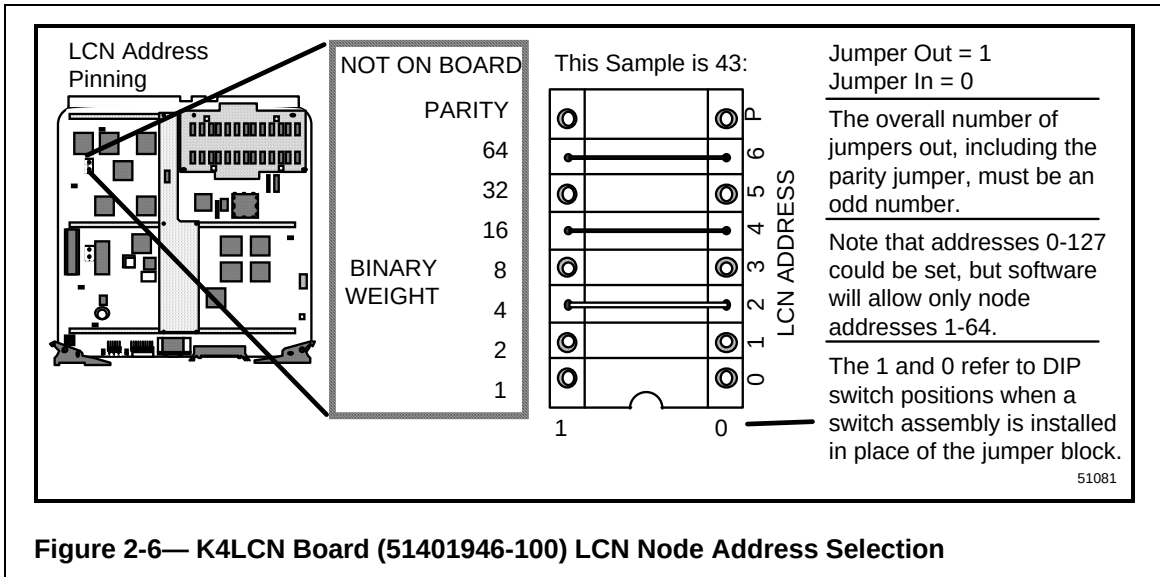
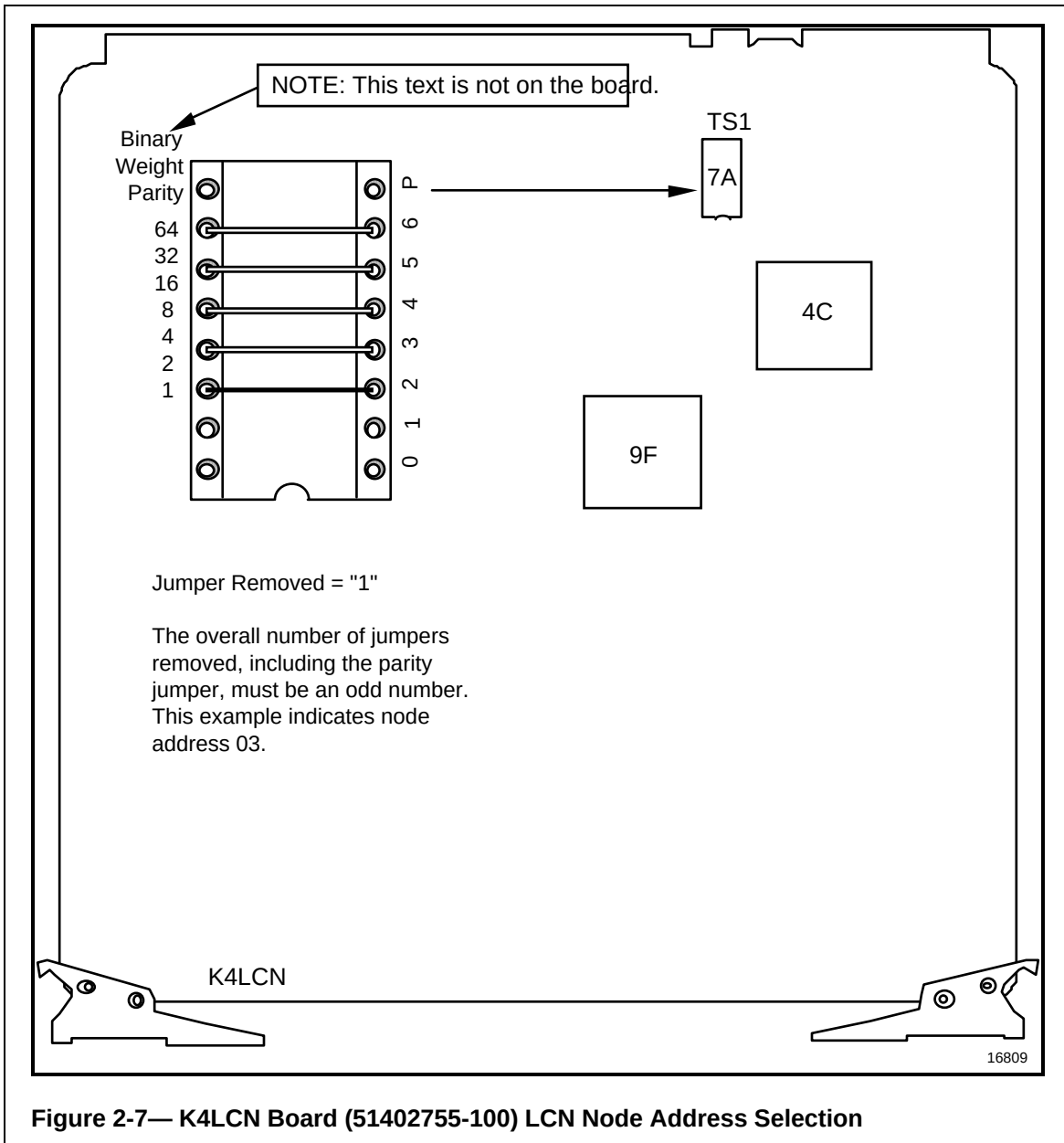


Figure 2-7 illustrates LCN node address selection on the K4LCN board, assembly 51402755-100. The pinning block, TS1, is located at coordinate 7A on the board. The absence of a jumper is a binary “1.” Parity must be an odd number.



The EPDGP I/O board can be pinned for default background shading as illustrated in Figure 2-8. The figure also illustrates the pinning for the vertical sync of the monitor.

Table 2-4 lists the EPDGP I/O assemblies available to match the type of Universal Station.

Universal Station Type	EPDGP I/O Assembly
Classic furniture US with keyboard power supply	51304584-100
Classic furniture US without keyboard power supply	51304584-200
Ergonomic furniture US with 21" FST monitor	51304584-300

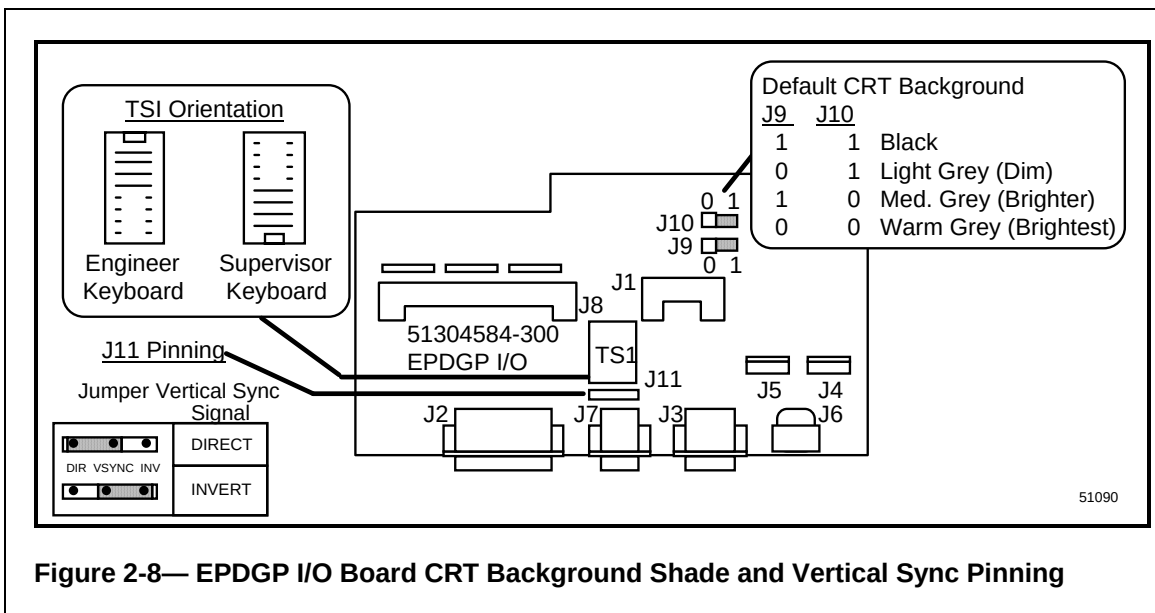


Figure 2-8— EPDGP I/O Board CRT Background Shade and Vertical Sync Pinning

Figure 2-9 illustrates the cabling to the EPDGP I/O board.

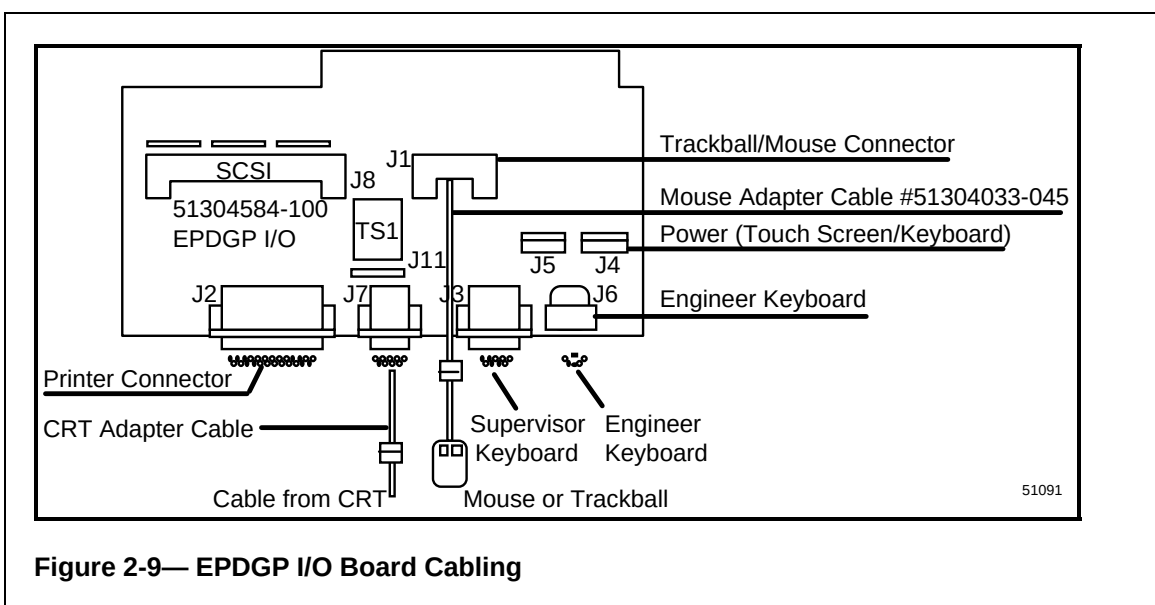
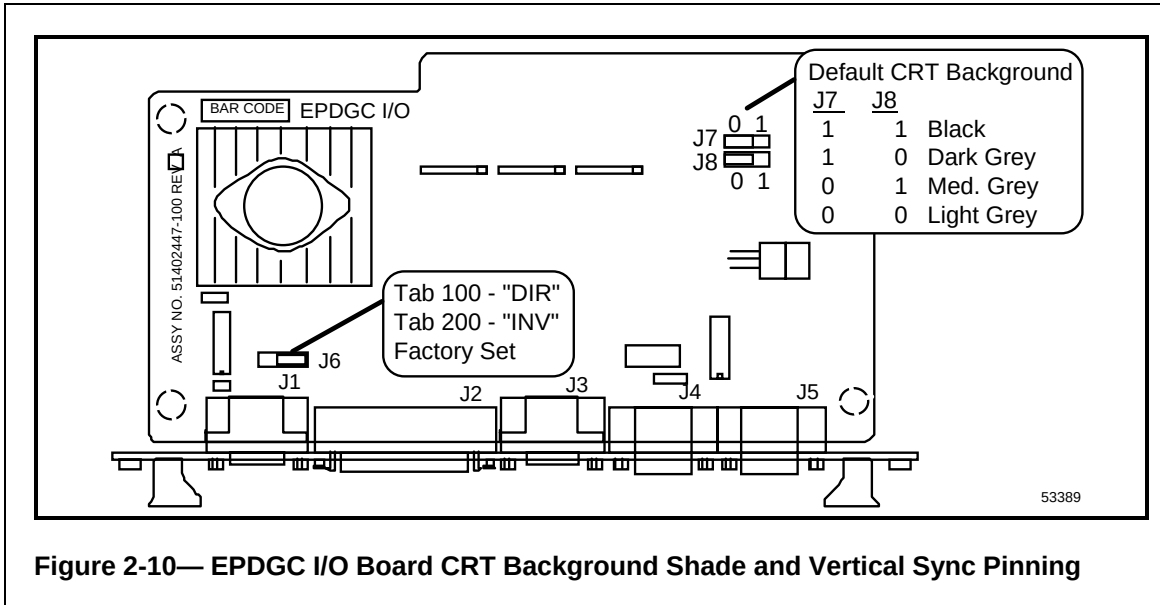


Figure 2-9— EPDGP I/O Board Cabling

The EPDGC I/O board can be pinned for default background shading as illustrated in Figure 2-10. The figure also illustrates the pinning for the vertical sync of the monitor.



2.8 LCN NETWORK GROUNDING

If the LCN coaxial cable bus segment does not have Clock Source/Repeater (CS/R) boards attached (an all-K2LCN and/or K4LCN segment), a grounding wire must be installed on one, and only one, CLCNA board and on one, and only one, CLCNB board. This grounds the shield of the A coaxial cable and the shield of the B coaxial cable at one point each to logic ground. Honeywell recommends that Cable A and Cable B not be grounded in the same module. This guards against loss of both LCN paths if the module is replaced.

Figures 2-11 and 2-12 illustrate the recommended ground connections in separate modules. If a coaxial cable segment has CS/R boards, the grounding of the cable sheaths is done at the CSRs and must not be duplicated at the CLCNA, CLCNB, or elsewhere.

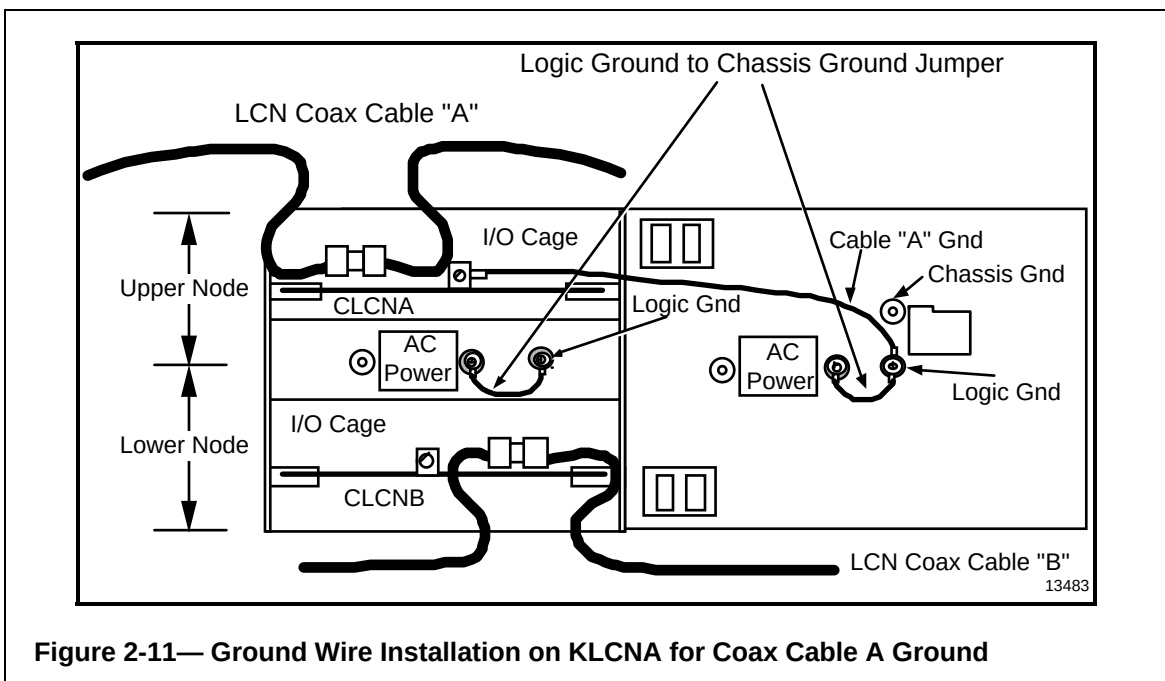
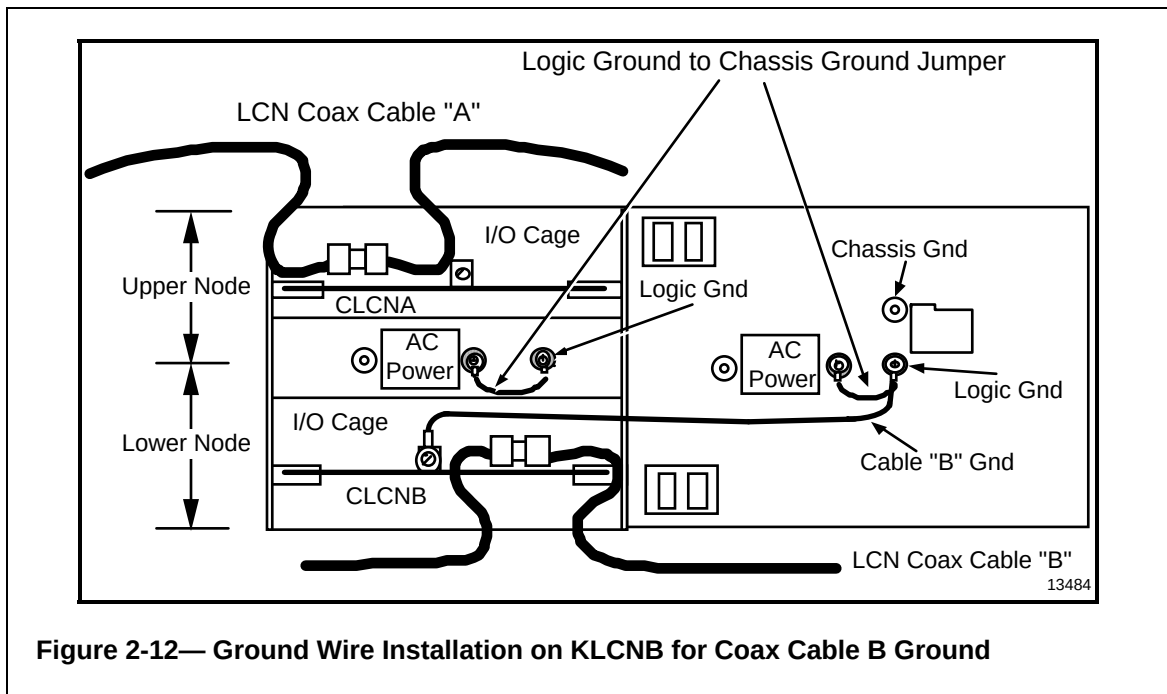


Figure 2-11— Ground Wire Installation on KLCNA for Coax Cable A Ground



2.9 PRECISION CLOCK PINNING (POWER SUPPLY REPLACEMENT)

The power supply has a pinning option for internal or external clock (see Figure 2-13). It is pinned for internal clock unless the Precision Clock Source option is installed.

If you are replacing the power supply, check the pinning on the old supply and ensure that the replacement is pinned the same.

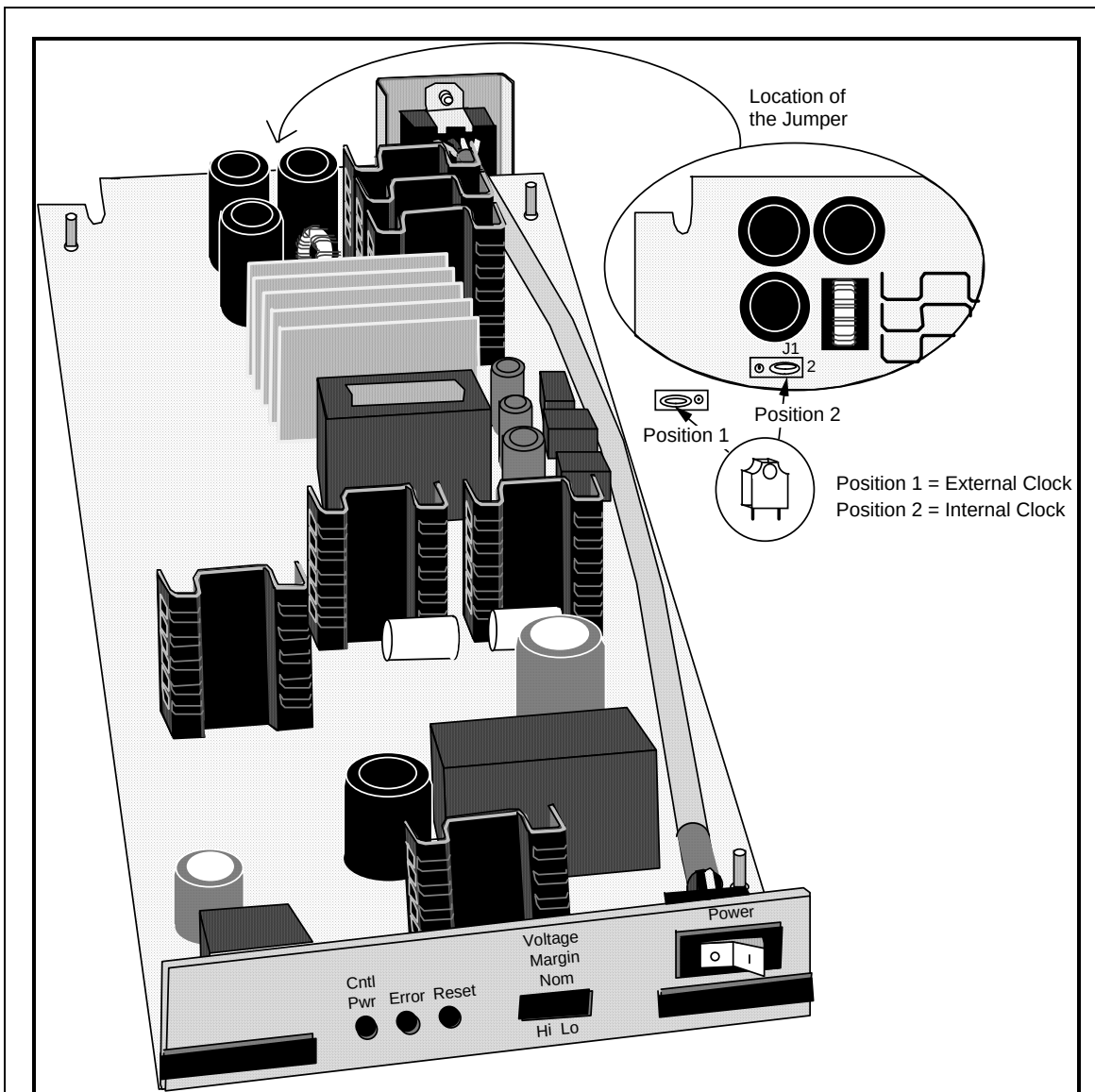


Figure 2-13— Power Supply Showing Location of Clock Jumper

4242

The CE Compliant power supplies come from two vendors. The external clock source pinning is illustrated in Figures 2-14 and 2-15.

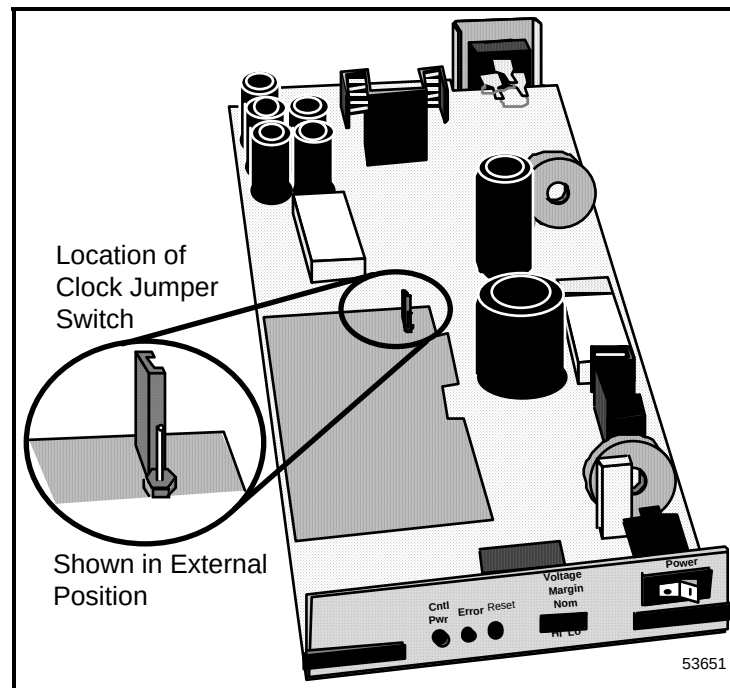


Figure 2-14— CE Compliant Power Supply Showing Location of Clock Jumper

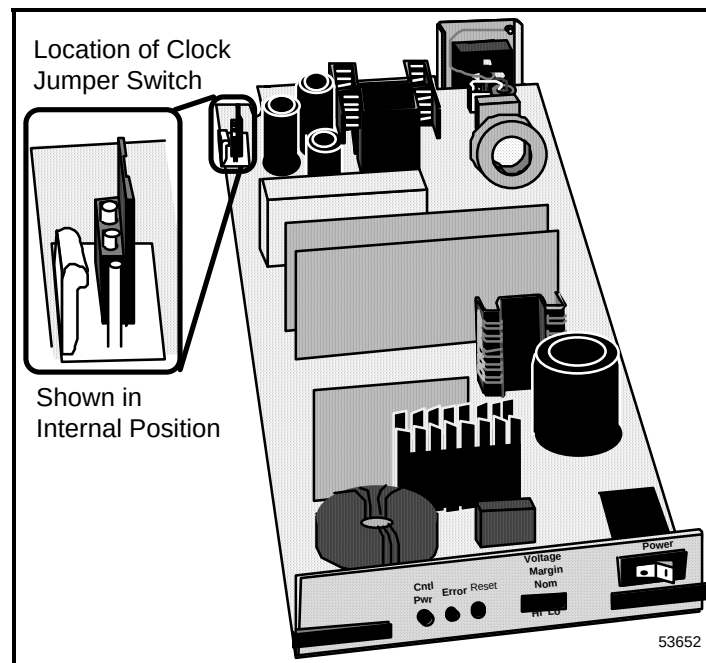


Figure 2-15— CE Compliant Power Supply Showing Location of Clock Jumper

2.10 NIM MODEM BOARD PINNING

Location SW1 has the revision pinning for the NIM Modem board. This pinning reflects the revision of the board and **is never to be changed unless the board is updated.**

Location SW2 has the pinning for the UCN node address. This address is a binary weighted number and is used only in the **Test** mode. The address entered, is the address of the primary NIM and must be an odd number with odd parity (an odd number of switches must be in the ON position). See Figure 2-16 for the non-CE Compliant NIM Modem board and Figure 2-17 for the CE Compliant NIM Modem board.

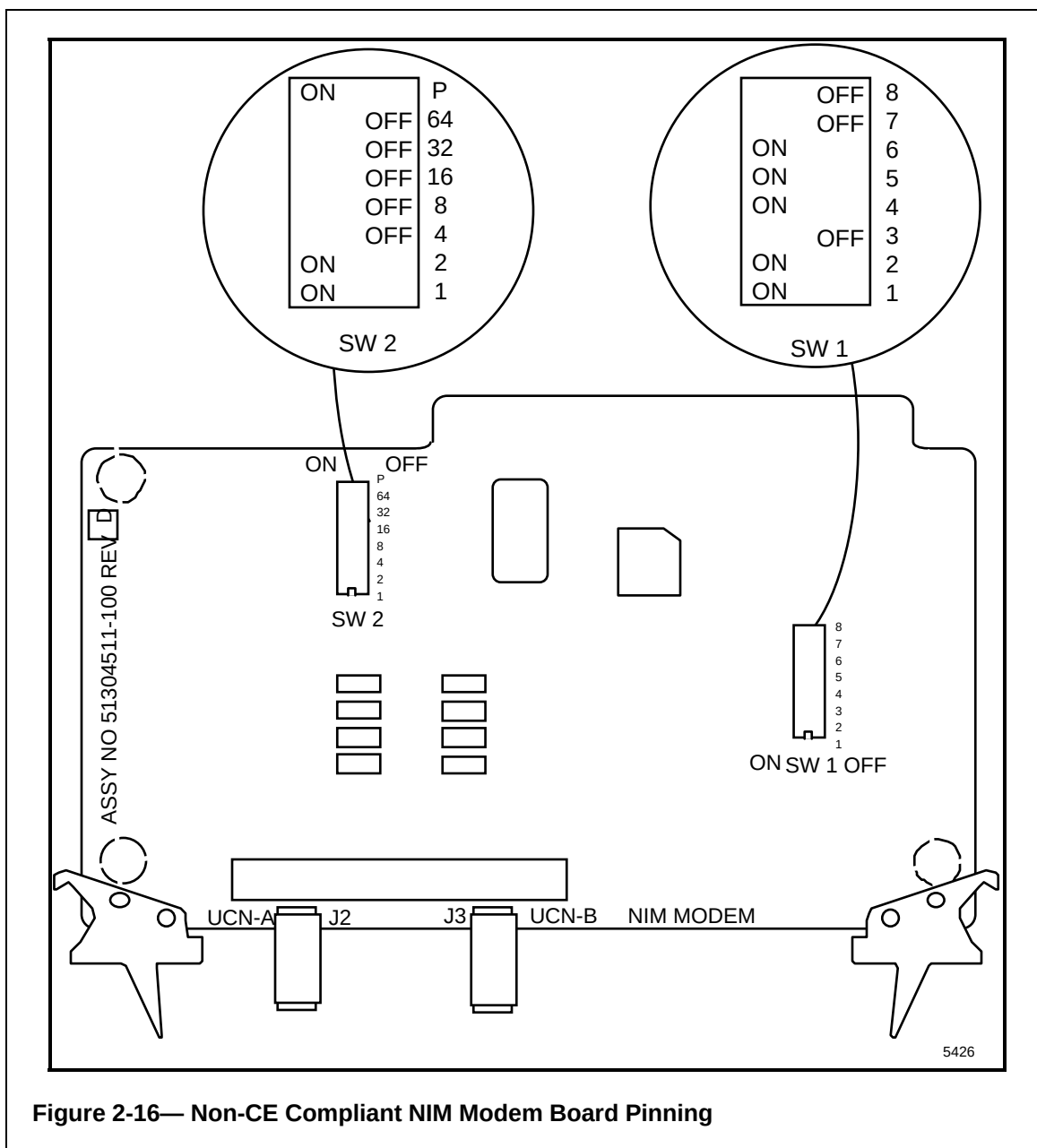
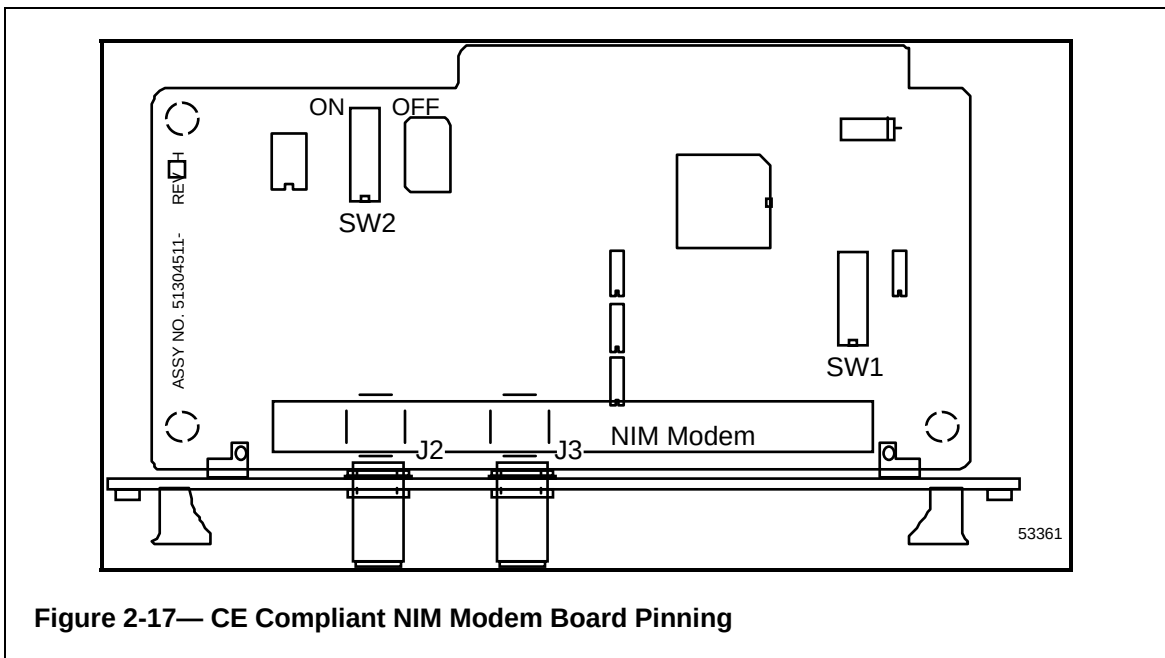


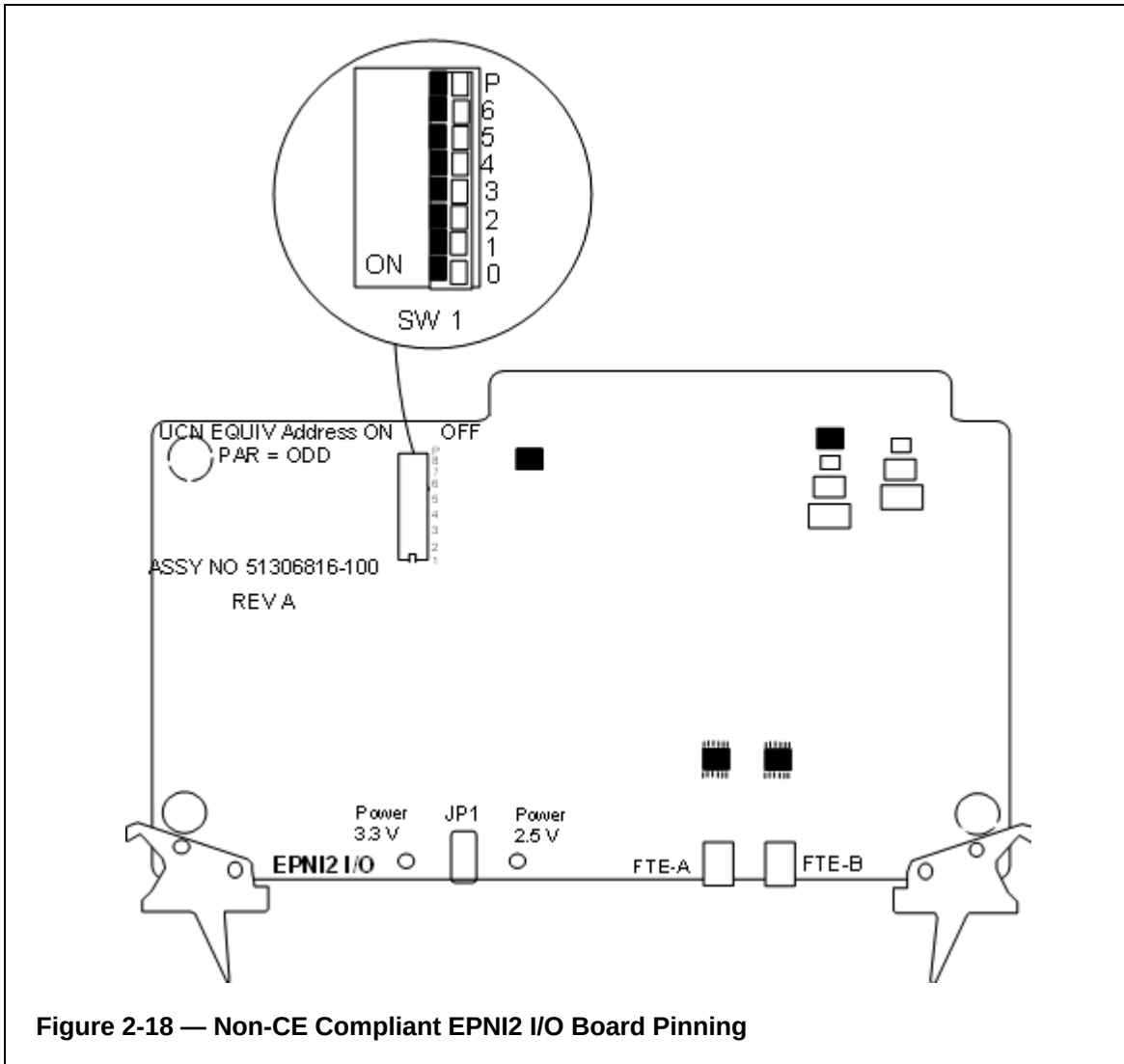
Figure 2-16— Non-CE Compliant NIM Modem Board Pinning

The CE Compliant NIM Modem board pinning is illustrated in Figure 2-17.



2.11 EPN2 I/O BOARD PINNING

Location SW1 has the pinning for the EUCN node address. This address is a binary weighted number and is used only in the Test mode. Enter the EUCN node address of the primary ENIM. It must be an odd number with odd parity (an odd number of switches must be in the OFF position). See Figure 2-18.



The CE Compliant EPNI2 I/O board pinning is illustrated in Figure 2-19.

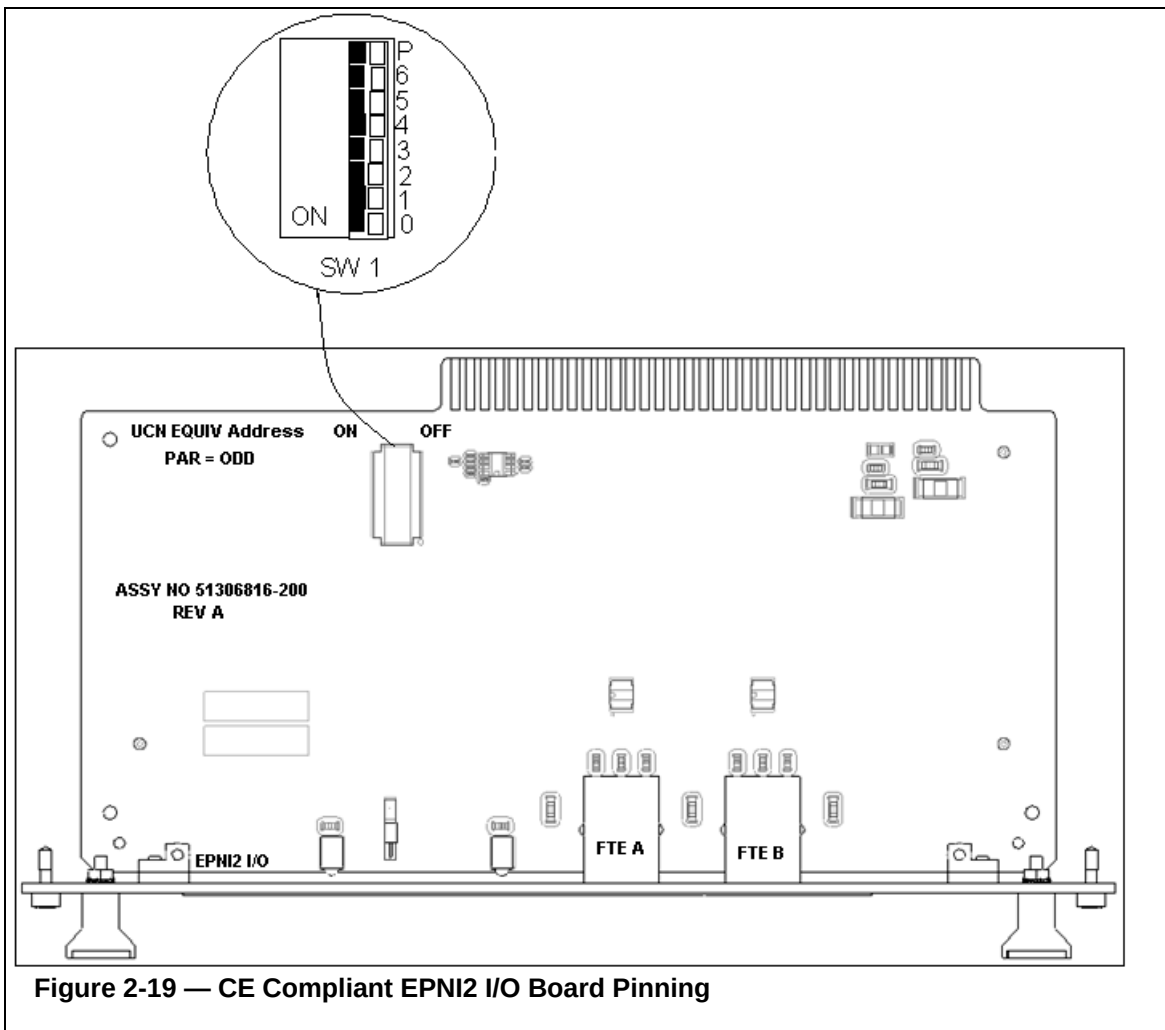


Figure 2-19 — CE Compliant EPNI2 I/O Board Pinning

EPNI2 I/O Board Pinning Example

Figure 2-20 illustrates the EUCN node address of 23. The figure shows the address weighting that each switch represents and switch 8 is for parity (odd). You add the binary count of each switch in the ON position to get the EUCN node address. Position the parity switch such that there are an odd number of switches in the ON position.

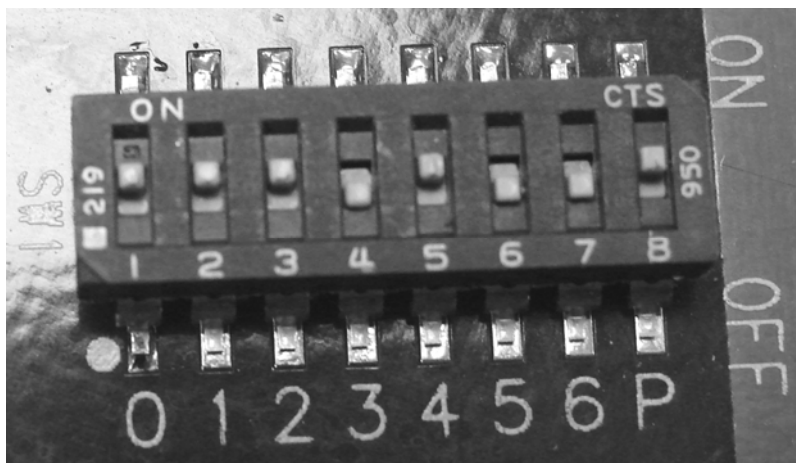


Figure 2-20 — Example: EPNI2 I/O Board Pinning

2.13 POWER SUPPLY LOADING IN LCNE/XLCNE2 HOST NODE

The LCN may be extended with one or more Fiber Optic Links. An otherwise empty node of a Dual Node Module may be used to host the XLCNE2 fiber optic transmit/receive boards. However, the XLCNE2 boards alone do not provide adequate load on the power supply to ensure that it will perform to specifications. A version of the KJMP board (51401594-200) with power supply loading resistors is used in this situation. Table 2-5 illustrates this configuration. The table indicates that more than one XLCNE2 could be installed in a node. While this is physically possible, system convention is to place the XLCNE2s on separate power supplies to prevent the loss of both Cable A and Cable B at the remote segment of the LCN if a power supply fails. If both nodes of a Dual Node Module are empty, each node could be used to host one of the two required XLCNE2s, because each node has a separate power supply.

NOTE

The XLCNE2 board is a replacement for the LCNE2 board. The two boards perform the same function although they use different connectors. The allowable fiber optic cable lengths allowed are also different for each board. An XLCNE2 can only talk to another XLCNE2, and the LCNE2 and XLCNE2 cannot be mixed on the same fiber.

Table 2-5— XLCNE2 Host Nodes

Lower Node			Upper Node		
Slot	Front	Rear	Slot	Front	Rear
3		Note	2	KJMP	Note
2	KJMP	Note	1		
1					

Note: Can be XLCNE2, FOC/XMTR, FOC/RCVR board or unused.

CAUTION

Do not insert a CS/R board in I/O Slot 1 of either node or serious damage can result.

TEST/TROUBLESHOOTING

Section 3

3.1 TESTS

Each node of the Dual Node Module is tested through firmware (in hardware) and software checks. Firmware tests built into the node provide two similar means of functionally checking the unit, whether or not it is connected to the LCN. The first firmware test starts when power is applied. Pressing the **RESET** button initiates a second (slightly different) firmware test.

Software tests start after the node is connected to the system. Loading the node personality, for example, initiates software Quality Logic Test (QLT).

Figure 3-1 is a flow diagram showing the relationship of the hardware and software tests.

3.2 TEST PROCEDURES

Each node must have a K2LCN or K4LCN board installed and must have access to a KLCNA (or CLCNA) and KLCNB (or CLCNB) boards on the RS-485 bus before it can be fully tested.

3.2.1 Power-On Test

Initiate power-on tests by pressing the “1” side of the power switch. Note that the red LEDs on the boards light for a few seconds (less than two minutes); then they should turn off and the green LEDs should turn on as the node passes its Quality Logic Tests. When the tests are complete, each node's alphanumeric display (on the K2LCN or K4LCN board) indicates its node address. Ensure that the expected node address is displayed.

Check the power supply **POWER STATUS** LEDs. Be sure that the green LED is on and that the red LED is off. Check the red **FAN ALARM** LED to ensure that it is not on or blinking.

3.2.2 Reset Test

If this is not a History Module, press the **RESET*** button. The red LEDs should go on momentarily as above, and then should go out when the green LEDs come on. Recheck the power supply, board, and fan LEDs as above. Note that each alphanumeric display indicates its node address.

If any of the above conditions does not pass, refer to subsection 3.3, *Troubleshooting*.

WARNING

* In the case of the History Module only, if you press the RESET button indiscriminately while the system is operating, you may cause damage to the data stored on the disk drives. For more details, see subsection 3.2, *Troubleshooting*, in the *History Module Service* manual.

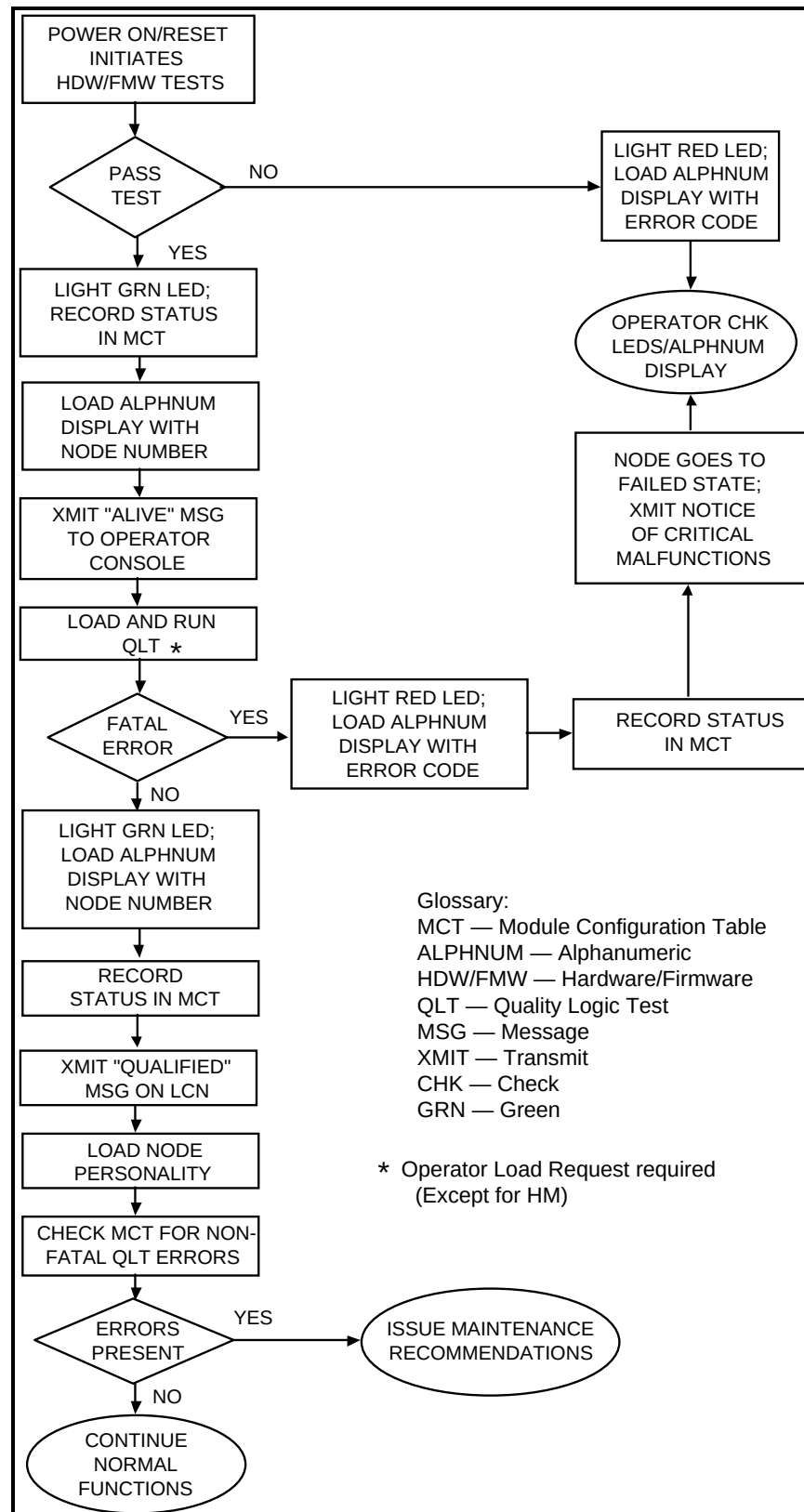


Figure 3-1 — Testing Flow Diagram

2898

3.2.3 Quality Logic Test

Load the node personality while observing the LEDs on the boards. The first part of the load process automatically loads and runs the Quality Logic Test. As the QLT runs, the red and green LEDs should cycle as they did during the power-on (firmware) test (see subsection 3.2.1). The QLT takes only a few seconds to run, whereas the firmware test takes considerably longer. When the QLT finishes, check that on all boards the green LED is on and no red LEDs are on. Check that the correct node address is displayed on the alphanumeric display on the K2LCN or K4LCN board.

3.3 TROUBLESHOOTING

The troubleshooting procedures in this manual are grouped by affected ORU and are keyed to the LED indicators and the alphanumeric display. First, a malfunction symptom is listed, then instructions are given to check or replace parts in order of the most likely causes. Perform the first indicated step, and then test. If the malfunction is still present, perform the next indicated step and test again, continuing in this manner until the malfunction is corrected. Do not perform the remaining steps after the malfunction has been corrected.

In the following procedures, the indicated LEDs are red unless otherwise stated. Refer to Section 4 for disassembly and assembly instructions.

The KLCNA (or CLCNA) and KLCNB (or CLCNB) boards were designed so that they can be replaced while power remains on. Also, the fan may be unplugged for a brief period for troubleshooting while power is on (see subsection 3.3.1). In all other cases, always turn power off before removing or replacing a part on the node, as this protects equipment from voltage transients and ensures proper initialization. You may turn power off for one node while leaving the other node on and operating. If you can't correct the fault by performing these procedures, refer to Appendix A for alphanumeric display troubleshooting. If you still cannot isolate the fault, contact the Honeywell Technical Assistance Center (TAC) in the U.S., or the equivalent service in your area. The telephone number for TAC is 1-800-822-7673 (1-602-863-5558 for Arizona callers).

CAUTION

All assemblies used in this module are likely to contain electrostatically sensitive devices. Use a personnel grounding strap and grounded work surfaces and equipment. Store and transport parts only in electrostatically safe containers.

The K2LCN or K4LCN board has indicators to assist in troubleshooting. The indicators are illustrated in Figures 3-2 and 3-3. There is a green LED indicator that indicates when the board has successfully completed its self-test program. There is a yellow LED indicator that indicates when data is being transmitted to the LCN network. There is a red LED bar display that indicates that an error was detected in the self-test program. Individual indicators in the bar display may blink during the self-test procedure. Successful completion of the self-test program is indicated when the green LED indicator is on and all of the red indicators in the bar display are off.

Figure 3-2 defines the indicators for early production K2LCN and K4LCN boards, assemblies 51401551-x00 and 51401946-100, respectively.

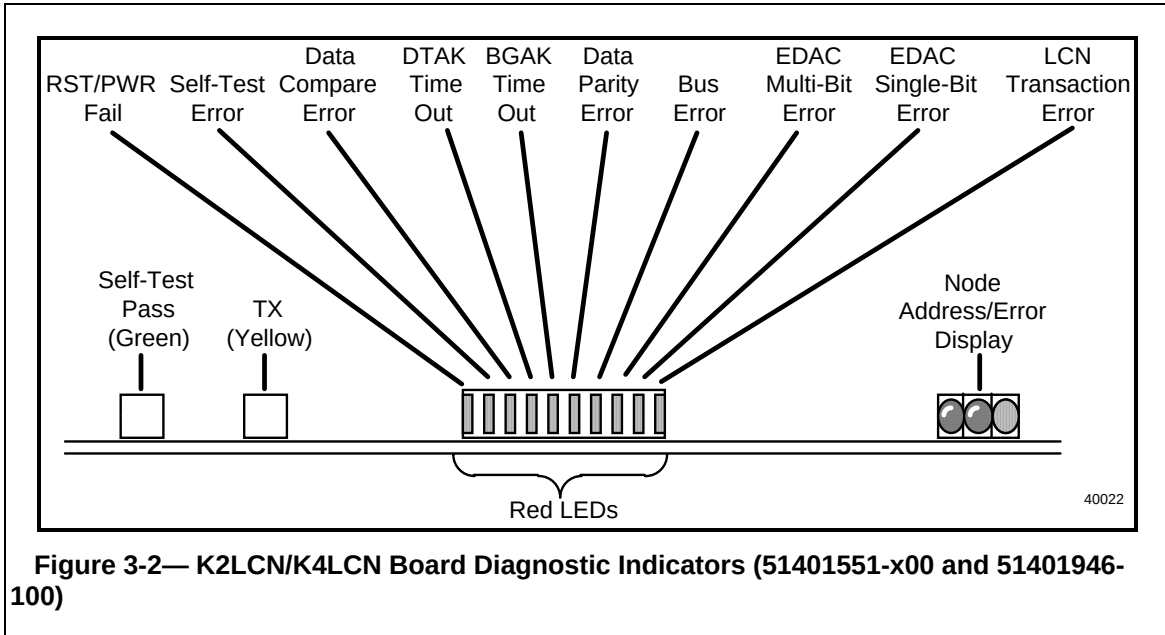
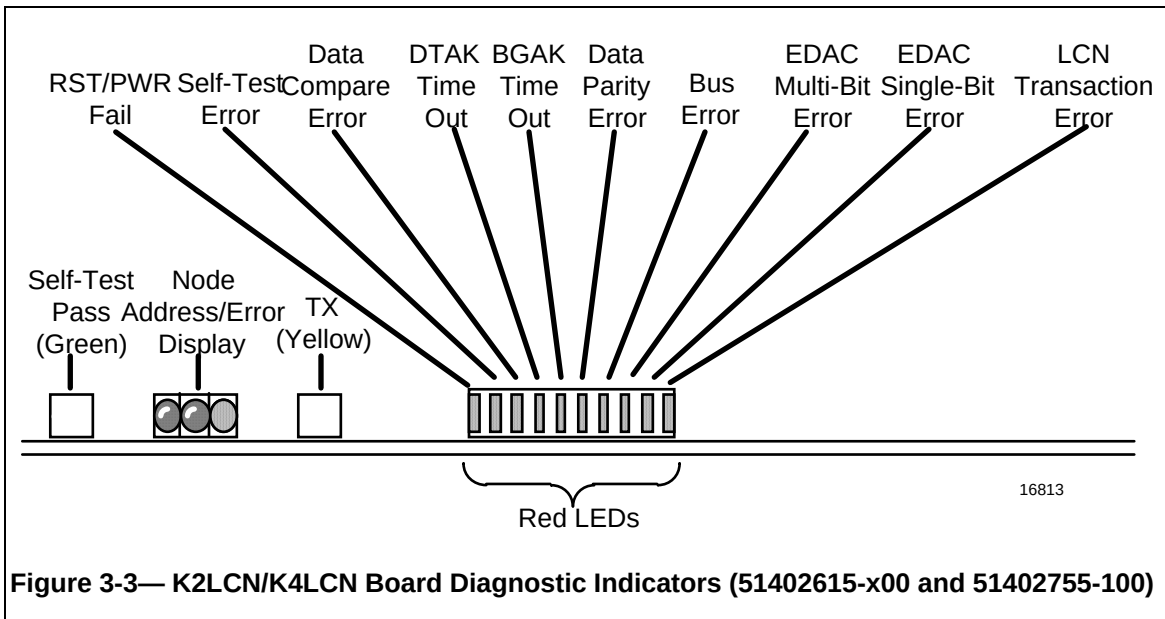


Figure 3-3 defines the indicators for later production K2LCN and K4LCN boards, assemblies 51402615-x00 and 51402755-100, respectively.



The following acronyms or abbreviations are used in Figures 3-2 and 3-3.

MOD	Module
LCN	Local Control Network
RST/PWR	Reset/Power
DTAK	Data Transfer Acknowledge
BGAK	Bus Grant Acknowledge
EDAC	Error Detection and Correction

The K2LCN and K4LCN boards also have a three-digit alphanumeric display. If the self-test program completes without failure, the display will show the LCN node address that has been pinned on the K2LCN or K4LCN board. If a failure occurs during the self-test program, the display will contain an error identification code preceded by a minus sign. Appendix A contains additional information on interpretation of the alphanumeric display.

3.3.1 Power Supply/Fan

No LED indicators are lit.

- Check that the power cord is plugged in and that the power switch is on. If they are, unplug the power cord and replace the power supply.

The red LED indicator on the Power Status panel is lit but, the **FAN ALARM** LED indicator is off.

- Unplug the power cord and replace the power supply.

The **FAN ALARM** LED indicator is lit.

WARNING

Keep fingers and other objects clear of fans. They have rotating parts and sharp fan blades.

The Fan Assembly in the Dual Node Module is wired to both power supply connectors. It will operate if either (or both) of the power supplies are on. Therefore, if the module contains two power supplies, you must turn both power supplies off in order to turn the Fan Assembly off.

- With power applied to the node, loosen the two quarter-turn bail-headed fasteners holding the fan assembly, and quickly slide the assembly out of the module. If the fans are functioning properly, they will remain turning for a few seconds. After the fans have stopped, check for bearing drag by attempting to rotate the blades with your fingers. In order to prevent overheating of the node, do not leave the fan assembly out of the module for than two minutes while performing this diagnosis.
- If either or both fans are not turning, or if the bearings are dragging (if there is any noticeable resistance to turning), replace the fan assembly. If you do not have a replacement fan assembly with you, reinstall the faulty unit while obtaining a replacement (assuming that the faulty unit is partially functional). This will provide some cooling.
- If the problem is still not corrected, check the fan power connector at the backpanel. If no problems are observed, replace the power supply.

3.3.2 Controller Board (FDC, CNI, DHI, CLI, SIO, SPC) Faults

SELF TST/ERR indicator does not go out.

- Ensure that the I/O board is properly installed.
- Replace the controller board.
- Replace the I/O board.

SELF TST/ERR indicator goes out, but the **PASS MOD TEST** (green) indicator does not light on a controller board.

- Replace the controller board and boards in the slots below it one at a time.
- Replace the K2LCN or K4LCN board.

SELF TST/ERR indicator goes out, but the **PASS MOD TEST** (green) indicator does not light on all controller boards.

- Replace the K2LCN or K4LCN board.
- Remove the top controller board, then apply power. Repeat for each controller board until the fault is cleared. Retry the extracted boards.

BUS TRAN ERR indicator is lit on the controller board and the **DTAK TIME OUT** indicator is lit on the K2LCN or K4LCN board.

- Replace controller board and the boards in slots below it one at a time.
- Replace the K2LCN or K4LCN board.

BUS TRAN ERR indicator is lit on a the controller board and the **BGAK TIME OUT** indicator is lit on the K2LCN or K4LCN board.

- Check for more than two open slots between the boards.
- Remove the top controller board, then apply power. Repeat for each controller board until the fault is cleared. Retry extracted the boards.
- Replace the K2LCN or K4LCN board.

3.3.3 PLCI Board Faults

NOTE

For additional information on troubleshooting the PLCI board, refer to the *PLC Gateway Planning, Installation, and Service* manual in the *Implementation/PLC Gateway* binder.

SELF TST/ERR indicator does not go out.

- Ensure that the PLCI I/O board is properly installed.
- Replace the PLCI board.
- Replace the PLCI I/O board.

SELF TST/ERR indicator goes out, but the **PASS MOD TEST** (green) indicator does not light on one controller board.

- Replace the PLCI board.
- Replace the K2LCN or K4LCN board.

SELF TST/ERR indicator goes out, but comes back on after the HG personality has been loaded.

- Incorrect software release or incorrect configuration. Refer to the *Programmable Logic Controller Gateway Planning, Installation, and Service* manual.
- Replace the PLCI board.
- Replace the K2LCN or K4LCN board.

BUS TRAN ERR indicator is lit on the PLCI board.

- Replace the PLCI board.
- Replace the K2LCN or K4LCN board.

3.3.4 Processor Board (K2LCN/K4LCN) Faults

DATA PAR ERR and **BUS ERR** are lighted on K2LCN/K4LCN.

- Check the alphanumeric display. The first two characters indicate the slot number of the failed board. The third character is the failed test number. Refer to Appendix A for specific alphanumeric codes.
- Replace the indicated board.
- Replace K2LCN or K4LCN board.

3.3.5 EPDG Board Faults

SELF TST/ERR indicator does not go out on the EPDG board.

- Replace the I/O card.
- Replace the EPDG board.
- Replace the K2LCN or K4LCN board.

SELF TST/ERR indicator goes out on the EPDG board, but the **PASS MOD TEST** (green) indicator does not light on the EPDG board.

- Replace the I/O card.
- Replace the EPDG board.
- Replace the K2LCN or K4LCN board.

BUS TRAN ERR indicator is lit on the EPDG board and the **DTAK TIME OUT** indicator is lit on the K2LCN or K4LCN board, but the **BUS ERR** indicator is not lit.

- Replace the I/O card.
- Replace the EPDG board.
- Replace the K2LCN or K4LCN board.

BUS TRAN ERR indicator is lit on the EPDG board and the **DTAK TIME OUT** and **BUS ERR** indicators are lit on the K2LCN or K4LCN board.

- Replace the I/O card.
- Replace the EPDG board.
- Replace the K2LCN or K4LCN board.

3.3.6 EPNI Board Faults

NOTE

The EPNI board is a later version of the PNI board.

The EPNI board is controller board which interfaces the bus and processor in a way that is similar to the controller boards listed in Section 3.3.2. First, check the **SELF TST/ERR** indicator (red; should be out) and the **PASS MOD TEST** indicator (green; should be on) on the EPNI board.

SELF TST/ERR indicator (red) is driven by a microprocessor on the EPNI board. If it is lit, check for the following causes:

- There was a hardware failure on the EPNI board.
- A problem was detected on-line (for instance, there may have been a EPNI local RAM parity error or a duplicate address may have been detected).
- The node was shut down (stunned) due to a watchdog timeout.
- The number of raw errors detected exceeded a pre-set threshold, causing the software on the EPNI board to enter the failed state.

If the state of the **SELF TST/ERR** indicator and **PASS MOD TEST** indicator are correct, continue with this instruction. Under normal system operating conditions, the following indicators and connections are present on the EPNI board.

- **Red LED** is out.
- **Green LED** is lit.
- **TX yellow LED** either blinks on and off (indicating heavy traffic) or stays on (heavy on).

Check that there are no disconnected or broken cables. If part of the UCN has failed, the failure reporting and diagnostic tests that are included in the software will help isolate the problem.

NOTE

ENIM differs from the NIM since it uses FTE cables that provide an IP Address. The ENIM node must receive an IP address to function properly. If there are no FTE cables connected, then the ENIM cannot be loaded.

Because of this difference between NIM and ENIM, the indication given by the left-most red LED on the front edge of the EPNI versus the EPNI2 board also differs. In the EPNI (NIM), if there are no coaxial UCN cables connected, the red LED light glows. In the EPNI2 (ENIM), the left-most red LED does not glow if there are no FTE cables connected. At least one FTE cable must be connected for the ENIM to receive an IP address.

3.3.7 EPNI2 Board Faults

NOTE

The EPNI2 board is a later version of the EPNI board.

The EPNI2 board performs a self-test. The self-test includes diagnostics to verify the operation of the I/O board, microprocessor kernel, and the support circuitry. In addition, it performs a loopback test of the data paths. The LEDs at the front of the EPNI2 board indicate the status.

Under normal circumstance the LEDs on the EPNI2 board displays the following:

LEDs	Status
SELF TST/ERR	Red is out
PASS MOD TEST	Green is on

SELF TST/ERR indicator (red) is driven by a microprocessor on the EPNI2 board. If the LED (red) is lit, check for the following causes:

- A hardware failure on the EPNI2 board during or after self-test.
- Any online problem (for instance, there may have been an EPNI2 local RAM parity error or duplicate address detection).
- The Node shut down (stunned) due to a watchdog timeout.
- The number of raw errors detected exceeded a pre-set threshold, causing the software on the EPNI2 board to enter the failed state.

Ensure that there are no disconnected or broken cables. If part of the EUCN fails, the failure reporting and diagnostic tests included in the software helps you to isolate the problem.

3.3.8 Fiber Optic Extender Board (XLCNE2, FOCT, FOCR, CS/R) Faults

NOTE

The XLCNE2 board is a replacement for the LCNE2 board. The two boards perform the same function although they use different connectors. The allowable fiber optic cable lengths allowed are also different for each board. An XLCNE2 can only talk to another XLCNE2, and the LCNE2 and XLCNE2 cannot be mixed on the same fiber.

3.3.7.1 Normal Indicators

Under normal system operating conditions, the following indicators and switch settings are present on the LCNE Extender Set.

- **Red LED** is out.
- **Green LED** is lit.
- **Yellow LED** is either blinking on and off (indicating traffic) or stays on (heavy traffic).
- **RUN/RESET** switch on the LCNE or XLCNE2 is in the run position.

3.3.7.2 Troubleshooting Indicators

If a remote segment has gone dead, check the following:

1. **Red LED on all LCNE or XLCNE2 boards should be off.** If a red LED is on, the Antijabber Flag is set. The failure that caused this to happen is on the board with the LED on. Replace that board using the procedures in the *LCN Guidelines* manual. See the “LCN Fiber Optic Link Service” section.
2. **Green LED on LCNE or XLCNE2, FOCR must be on steady.** If a green LED is not lit (or is flickering), there is not enough light power being received from the fiber optic cable. Check the following:
 - a. Cable integrity: Check for cut, damaged, or kinked cable.
 - b. Light received: Measure the light power at the receiving end of the cable. A minimum of 2.5 μ W must be measured.
 - c. Light transmitted: Measure the light power at the transmitter, a minimum of 50 μ W must be measured, using a one-meter length of fiber optic cable.

NOTE

If the Green LED is on continuously, there may still not be enough light power being received from the fiber optic cable. The Green LED is not an indication of signal quality. The Green LED indicates the presence of an attached cable.

3. **Yellow indicator should be on as data traffic is passed.**

If a fiber optic connection that had previously worked ceases to do so, and no fault can be found with the LCNE/XLCNE2/FOCT/FOCR boards, make the power measurements discussed in the *LCN Guidelines* manual. See the “Fiber Optic Light Power Measurements” section.

3.3.9 RS-485 to LCN Converter Boards CLCNA (KLCNA), CLCNB (KLCNB)

For information on troubleshooting LCN problems, see Section 4 in *LCN Guidelines, Implementation, Troubleshooting, and Service* manual.

If replacement of a CLCNA or CLCNB board is indicated, the following points are important to know. Read all of the following points before beginning the replacement procedure.

- If the CLCNA (KLCNA) or CLCNB (KLCNB) board being replaced is one which has a coaxial cable grounding wire on it (refer to Figure 2-6), use the following procedure. Before disconnecting the LCN coaxial cable from the CLCNA (KLCNA) or CLCNB (KLCNB) board, connect one end of a clip lead to a metal part of the BNC Tee-connector and connect the other end of the clip lead to Logic Ground. This procedure maintains LCN cable ground during the board replacement, allowing communications to continue over the cable.
- Before removing the CLCNA (KLCNA) or CLCNB (KLCNB) board, disconnect the LCN cable by disconnecting the BNC tee from the coaxial connector on the CLCNA (KLCNA) or CLCNB (KLCNB) board. Do not remove the individual cables from the tee—they must remain connected so that communications can continue among the other nodes.
- The CLCNA (KLCNA) and CLCNB (KLCNB) boards were designed to be plugged and unplugged from the I/O card cage with power on. Therefore, it is not necessary to turn off power to the node during CLCNA (KLCNA) or CLCNB (KLCNB) board replacement. **DO NOT** do this with any other board type or substantial damage may occur.
- While the BNC tee is disconnected from the CLCNA (KLCNA) or CLCNB (KLCNB) board, remember that there are exposed metal parts on this Tee-connector (and the clip lead if present), and these parts are at ground potential. Do not allow these metal parts to contact other components, or errors or damage may result.

The CLCNA and CLCNB board assemblies have replaced the KLCNA and KLCNB boards in current production and provide the same functionality.

CAUTION

KLCNA or KLCNB boards must not be mixed with CLCNA or CLCNB boards in the same Dual Node chassis.

3.3.10 NGI Board

SELF TST/ERR indicator does not go out.

- Ensure that the NGIO board is properly installed.
- Replace the NGI board.
- Replace the NGIO board.

SELF TST/ERR indicator goes out, but **PASS MOD TEST** (green) indicator does not light on one controller.

- Replace the NGI board, and then the other NGI board in slot below it, if present (three-board, redundant cable node).
- Replace the K2LCN or K4LCN board.

SELF TST/ERR indicator goes out, but the **PASS MOD TEST** (green) indicator does not light on all controllers.

- Replace the K2LCN or K4LCN board.
- Replace the NGI board, then apply power. Repeat for the other NGI board, if present. Retry the extracted boards.

BUS TRAN ERR indicator is lit on the controller, and the **DTAK TIME OUT** is lit on the K2LCN or K4LCN board.

- Replace the NGI board, and then apply power. Repeat for the other NGI board, if present. Retry the extracted boards.
- Replace the K2LCN or K4LCN board.

BUS TRAN ERR indicator is lit on a controller, and **BGAK TIME OUT** is lit on the K2LCN or K4LCN board.

- Replace the NGI board, and then apply power. Repeat for the other NGI board, if present. Retry the extracted boards.
- Replace the K2LCN or K4LCN board.

The **TX** (yellow) indicator indicates transmit activity. If this indicator is not on, troubleshoot the Plant Information Network (PIN) cables and other interfaces.

CAUTION

There is a test jumper at the front of the NGI board (near the indicator lamps that are described above). This jumper has two positions, **TEST** and **NORMAL**. The jumper should always be in the **NORMAL** position when the network is operating. Placing this jumper in the **TEST** position can disrupt operation of other nodes operating on the PIN.

DISASSEMBLY/ASSEMBLY

Section 4

4.1 DISASSEMBLY

WARNING

To protect equipment from voltage transients and ensure reliable operation, always turn power off before removing or replacing boards in a node. While working on one node in a Dual Node Module, you can leave the power on for the other node, leaving it operational. It is **NOT** necessary to turn off power to replace a KLCNA (CLCNA) or KLCNB (CLCNB) board. See subsection 3.3.8.

Disassemble the module only to the extent necessary to service or replace defective parts. The following procedures may require the use of a blade-tip screwdriver, nut drivers, a personnel grounding strap, and antistatic fixtures and equipment. Item numbers enclosed in parentheses in the procedures below refer to circled numbers in the exploded module diagram, Figure 4-1, and are keyed to the item numbers in *Spare Parts*, Section 5.

1. Remove the front panel (1) by rotating the two quarter-turn fasteners on the right side of the panel. Gently pull the right edge of the front panel away from the chassis approximately one inch and then slide the panel to the right, approximately one-half inch, until the panel is free. Do not attempt to swing the panel open like a door, or damage will result.

CAUTION

All assemblies used in this module are likely to contain electrostatically sensitive devices. Use a personnel grounding strap and grounded work surfaces and equipment. Store and transport parts only in electrostatically safe containers.

2. Remove the boards (2) by pulling outward on the retainers and sliding the board out of the chassis.
3. Remove the fan assembly (3) by loosening the two quarter-turn bail-headed fasteners and sliding the assembly out of the module.
4. Remove the power supply (4) by pulling the retainers, and sliding the supply out of the chassis.
5. Remove the I/O boards (5) from the rear chassis.

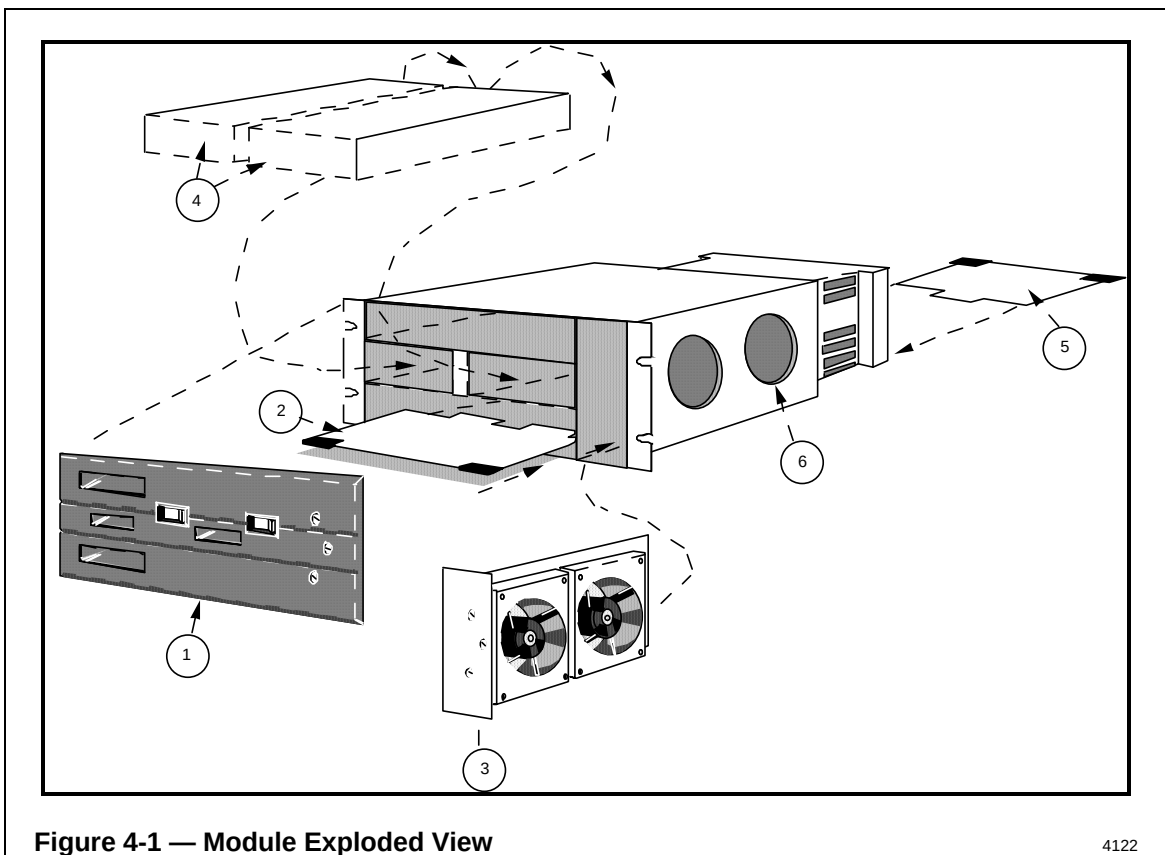


Figure 4-1 — Module Exploded View

4122

NOTE

Before inserting a replacement board:

1. Be sure that the replacement board is at a revision level equal to, or later than, the board being replaced.
2. Be certain that any pins or switch settings (e.g., the node address on the K2LCN or K4LCN board) matches the settings on the board being replaced. See subsection 2.7 for additional information on pinning options.

4.2 ASSEMBLY

Perform the following procedure to assemble the Dual Node Module.

1. Insert the I/O boards (5) in the rear chassis, taking care not to damage connectors, and secure the board retainers. Be sure that the I/O boards match the primary boards.
2. Insert the power supply (4) into the chassis and secure the retainers.
3. Insert the fan assembly (3) into the chassis and secure the bail-headed screws.
4. Insert the boards (2) into the chassis and secure the retainers.
5. Attach the front panel (1) to the chassis (6) and secure the quarter-turn fasteners.

SPARE PARTS

Section 5

5.1 INTRODUCTION

Table 5-1 lists spare parts for the module. Those parts at the ORU level are marked with an asterisk (*). The listing is arranged in item-number order with the items keyed to the circled numbers in Figure 4-1. Also listed are those boards required for the various Dual Node Module configurations, referenced in Tables 2-2 and 2-3. In some cases more than one part is assigned to an item number (for instance, any one of several board assemblies can occupy a slot in the module). In this case, the list is in numerical order by part number.

5.2 SPARE PARTS LIST

Table 5-1— Parts List

Item	Description	Part Number
1	Dual Node Module Front Panel Assembly	* 51401486-100
2	CLI Board Assembly, Computer/LCN Interface	* 80360206-001
	CNI Board Assembly, Computer Network Interface	* 51401088-100
	DHI Board Assembly, Data Hiway Interface	* 51400700-100
	EAMR, Enhanced Application Module Redundancy	* 51401996-100
	EPDG Board Assembly, Enhanced Peripheral Display Gen.	* 51401286-100
	EPNI Board Assembly, Enhanced Network Interface	* 51401583-100
	EPNI2 Board Assembly, Enhanced Network Interface	* 51454196-100
	FDC Board Assembly, Floppy Disk Controller	* 51400669-100
	K2LCN Board	
	With 2 megabytes of memory	* 51402615-200
	With 4 megabytes of memory	* 51402615-400
	With 8 megabytes of memory	* 51402615-800
	K4LCN Board without Daughter (Mezzanine) Memory Board	* 51401946-100
	4 Megabyte Daughter (Mezzanine) Memory Board	* 51201645-400
	8 Megabyte Daughter (Mezzanine) Memory Board	* 51201645-800
	16 Megabyte Daughter (Mezzanine) Memory Board	* 51201759-160
	K4LCN Board without DIMM Memory Board	* 51402755-100
	4 Megabyte DIMM Memory Board	* 51201795-400
	8 Megabyte DIMM Memory Board	* 51201795-800
	16 Megabyte DIMM Memory Board	* 51201793-160
	KJMP Board, Power Supply Load Application	* 51401594-200
	NGI Board Assembly, Network Gateway Interface	* 51401583-200

(Continued)

Table 5-1— Parts List (Continued)

Item	Description	Part Number
	PLCI Board Assembly, Programmable Logic Ctrl. Gateway	* 51400997-100
	PNI Board Assembly, Network Interface Module	* 51400955-100
	SIO Board Assembly, System Input/Output	* 51400855-100
	SPC Board Assembly, Smart Peripheral Interface Cont.	* 51401052-100
3	Dual Node Fan Assembly	* 51401496-100
4	Dual Node Power Supply	* 51401497-100
	LCN Power Supply Replacement Kit (includes new power cord)—New CE Compliant Power Supply for all modules Dual Node, Classic Furniture 120 Vac Dual Node, Classic Furniture 1240 Vac Dual Node, Classic Furniture 120/240 Vac	* 51402184-100 * 51402184-200 * 51402184-300
5	CLCNA, LCN I/O Board (Dual Node LCN Coax A I/O Board) EC	* 51305072-200
	CLCNA, LCN I/O Board (Dual Node LCN Coax A I/O Board) Replaces KLCNCA	* 51305072-500
	CLCNA, LCN I/O Board (Dual Node LCN Coax A I/O Board) EC	* 51305072-800
	CLCNA/B, LCN I/O Board	* 51305072-400
	CLCNA/B, LCN I/O Board EC	* 51305072-700
	CLCNB, LCN I/O Board (Dual Node LCN Coax B I/O Board) EC	* 51305072-300
	CLCNB, LCN I/O Board (Dual Node LCN Coax B I/O Board) Replaces KLCNCB	* 51305072-600
	CLCNB, LCN I/O Board (Dual Node LCN Coax B I/O Board) EC	* 51305072-900
	CLI I/O, Communications Line Interface I/O (RS-232C)	* 80360209-001
	CLI I/O, Communications Line Interface I/O (RS-449)	* 80360230-001
	CLI/A, Communications Line Interface (RS-449) EC	* 51196701-100
	CNI I/O, Communications Network Interface	* 51304537-100
	CNI, Communications Network Interface I/O Board EC	* 51304537-200
	Data Hiway Interface I/O	* 51108088-100
	DHI, Data Hiway Interface I/O Board EC	* 51108088-200
	EPDG I/O, Enhanced Peripheral Display Generator I/O	* 51304270-100
	EPDGC-1, Enhanced Peripheral Display Generator I/O Board- 1 EC	* 51402447-100
	EPDGC-2, Enhanced Peripheral Display Generator I/O Board- 2 EC	* 51402447-200

(Continued)

Table 5-1— Parts List (Continued)

Item	Description	Part Number
	EPDGP I/O, Enhanced Peripheral Display Generator I/O (replaces EPDG I/O)	* 51304584-300
	EPLCI I/O, Enhanced Programmable Logic Controller Interface I/O	* 51304812-100
	EPLCI I/O, Enhanced Programmable Logic Controller Interface I/O EC	* 51304812-200
	EPNI2 I/O, Enhanced Network Interface I/O	* 51306816-100
	EPNI2 I/O, Enhanced Network Interface I/O CE	* 51306816-200
	FDC I/O, Floppy Disk Controller I/O	* 51109336-100
	XLCNE2 Multi Mode (8Km)	* 51305508-100
	XLCNE2 Multi Mode (8Km) CE	* 51305508-200
	XLCNE2 Single Mode (20Km)	* 51305508-300 ¹
	XLCNE2 Single Mode (20Km) CE	* 51305508-400 ¹
	¹ For systems with XLCNE2s (Single Mode), If the Total Network Length exceeds 10.3 km (for a single remote segment topology) or 9.0 km (for a multiple remote segment topology), modifications to every LCN interface in the LCN system is required. That is, every LLCN board needs to be replaced by an XLLCN board and every K2LCN/K4LCN board needs to be replaced with an XK2LCN/XK4LCN board and for GUS stations the LCNP4 needs to be replaced with the XLCNP4.	
	NG FOM, Network Gateway Fiber Optic Modem EC	* 51306959-200
	NGIO, Network Gateway I/O Board	* 51304472-100
	NIM MODEM, NIM Modem I/O board (put in PNI I/O slot)	* 51304511-100
	NIM Modem, NIM Modem I/O EC	* 51304511-200
	PLCG Relay, Programmable Logic Controller Gateway Relay Board	* 51304421-100
	PLCG Relay, Programmable Logic Controller Gateway Relay Board EC	* 51304421-200
	PLCI I/O, Programmable Logic Controller Interface I/O	* 51195096-100
	PLCI I/O, Programmable Logic Controller Interface I/O Board EC	51195096-200
	PNI I/O, Process Network Interface I/O	* 51303944-100
	PNM I/O, Process Network Modem I/O	* 51303947-100
	SIO I/O, Serial Input Output Interface I/O	* 51108843-100
	SIOI-OE, Serial Input Output I/O Enhanced	* 51304814-100
	SIOI-OE, Serial Input Output I/O Enhanced EC	* 51304814-200
	SPC I/O, Smart Peripheral Controller I/O	* 51304156-100

(Continued)

Table 5-1— Parts List (Continued)

Item	Description	Part Number
	SPC-2, Smart Peripheral Controller I/O Board	* 51304907-100
	SPC-2, Smart Peripheral Controller I/O Board EC	* 51304907-200
	SPC-3 I/O, Smart Peripheral Controller I/O Board EC	* 51305088-100
	TP485 LCN Interface Board	* 51304776-100
	TP485 LCN Interface Board	* 51304776-200
	TP485-3, TP485 LCN Interface Board EC	* 51304776-300
	TP485-4, TP485 LCN Interface Board EC	* 51304776-400
	WDI, Winchester Drive Interface EC	* 51304903-200
	WDI, Winchester Drive Interface	* 51304903-100
6	Dual Node Module Assembly for Classic Furniture	* 51195752-100
	Dual Node Module Assembly for Ergonomic Furniture	* 51403144-100
	Dual Node Module Assembly for Classic Furniture (CE Compliant)	* 51196727-100
	Dual Node Module Assembly for Ergonomic Furniture (CE Compliant)	* 51196728-100
MISC.	Coax Terminator	30732052-001
	Relay, Solid State (Used on PLCG Relay Switch Board)	* 51190516-100
	Relay, DC Gen. Purpose (Used on PLCG Relay Switch Board)	* 51190526-100
	BNC Tee-Connector	51190728-105
	RS-485 Terminator Assembly	51195511-100
	Ground Cable (Logic to Chassis Ground)	51195754-100
	Ground Cable (LCN Cable Ground)	51195791-100
	PLCG Relay Switch Board Assembly	* 51304421-100
	RS-232 Jumper Cable	51304694-100

STARTUP Section 6

6.1 VISUAL CHECKS

- Check the Module power and signal cables to ensure that they are present and tight.
- Check the ground strap on rear of chassis for good connection.
- Ensure that all boards and power supply(s) are fully inserted into the connectors and that board retainers are secured.

6.2 INITIALIZE NODE

- Turn the power on by pressing the “1” side of the power switch. After 2 minutes or less, all red LEDs should be off and all green LEDs should be on, and the alphanumeric display should indicate the address of the node. If not, check indicators and LEDs in accordance with subsection 3.2, *Test Procedures*.
- If this is not a History Module, press the **RESET*** button. Check displays and LEDs as above.

WARNING

* In the case of the History Module only, if you press the RESET button indiscriminately while the system is operating, you may cause damage to the data stored on the hard disk. See subsection 3.2, *Troubleshooting*, in the *History Module Service* manual for complete details.

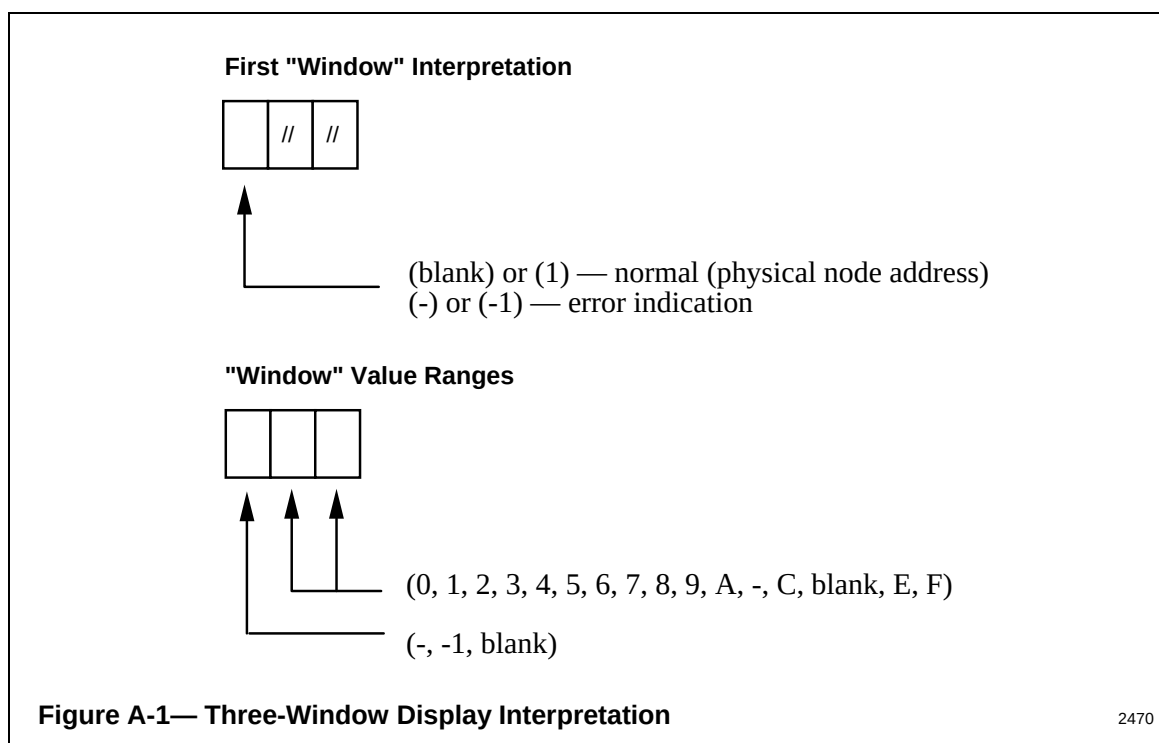
NOTE

When you replace or restart a board or module with an outstanding maintenance recommendation, be sure to make an SMCC corrective action entry. Otherwise, the system will not start a recount of errors. See the *System Maintenance Guide* manual for additional information about this procedure.

ALPHANUMERIC DISPLAYS

Appendix A

The alphanumeric display aids troubleshooting by displaying codes provided by firmware and software test programs. In interpreting the codes, pay special attention to the left-most character. As shown in Figure A-1, this character can have four distinct states, and reading the display correctly is critical to interpreting the code.



NOTE

Section 4 of the *Messages Directory* manual contains additional information on some of the error codes.

CAUTION

The error code interpretations that follow do not apply while the node is executing test software such as HVTS.

A.1 Recommended Actions for Specific Code Occurrences

Please note, when multiple recommendations are given for a window code-value, they are organized in order from most to least likely.

Recommendations marked with * indicate an unimplemented code-value. Any such occurrence should be brought to the attention of Honeywell TAC.

Window values marked as

//

 are “don’t care” values.

-	0	//
---	---	----

Replace the K2LCN or K4LCN board.*

-	1	n
---	---	---

n=(-) Replace the K2LCN or K4LCN board; replace the boards in slots 2 and 3 in turn if present.

n=(1 thru 8) Replace the K2LCN or K4LCN board.

n=(other) Replace the K2LCN or K4LCN board.*

-	2	N
---	---	---

n=(1 thru 7) Replace the board in slot 2; replace the K2LCN or K4LCN board; replace the module backplane or chassis assembly.

n=(-) or (C) Replace the board in slot 2; replace the K2LCN or K4LCN board; replace the board in slot 3 if present; replace the module backplane or chassis assembly.

n=(other) Replace the K2LCN or K4LCN board.*

-	3	n
---	---	---

n=(1 thru 7),
(-) or (C) Replace the board in slot 3 if present; replace the K2LCN or K4LCN board; replace the module backplane or chassis assembly.

n=(other) Replace the K2LCN or K4LCN board.*

-	4	n
---	---	---

n=(1 thru 7),
(-) or (C) Replace the K2LCN or K4LCN board; replace the module backplane or chassis assembly.

n=(other) Replace the K2LCN or K4LCN board.*

-	5	n
---	---	---

n=(1 thru 7),
(-) or (C) Replace the K2LCN or K4LCN board; replace the module backplane or chassis assembly.

n=(other) Replace the K2LCN or K4LCN board.*

-	6	n
---	---	---

n=(1 thru 7),
(-) or (C) Replace the K2LCN or K4LCN board; replace the module backplane or chassis assembly.

n=(other) Replace the K2LCN or K4LCN board.*

-	7	n
---	---	---

n=(1 thru 7),
(-) or (C) Replace the K2LCN or K4LCN board; replace the module backplane or chassis assembly.

n=(other) Replace the K2LCN or K4LCN board.*

-	8	n
---	---	---

n=(1 thru 7),
(-) or (C) Replace the K2LCN or K4LCN board; replace the module backplane or chassis assembly.

n=(other) Replace the K2LCN or K4LCN board.*

-	9	n
---	---	---

n=(1 thru 7),
(-) or (C) Replace the K2LCN or K4LCN board; replace the module backplane or chassis assembly.

n=(other) Replace the K2LCN or K4LCN board.*

-	A	n
---	---	---

n=(A) or (C) Replace the K2LCN or K4LCN board.

n=(other) Replace the K2LCN or K4LCN board.*

-	-	//
---	---	----

Replace the K2LCN or K4LCN board.*

-	C	//
---	---	----

Replace the K2LCN or K4LCN board.*

-		n
---	--	---

n=(A) If value persists after 30 seconds, replace the K2LCN or K4LCN board.

n=(other) Replace the K2LCN or K4LCN board.*

-	E	n
---	---	---

- n=(0) An unexpected interrupt has occurred - replace the K2LCN or K4LCN board.
- n=(1) Replace the K2LCN or K4LCN board; replace the boards in slots 2 and 3 in turn.
- n=(2) No working DRAM - replace the K2LCN or K4LCN board.
- n=(3) Suspected program error, - replace the K2LCN or K4LCN board.
- n=(5) Unverified cause of Level 7 interrupt, - replace the K2LCN or K4LCN board; replace the power supply.
- n=(other) Replace the K2LCN or K4LCN board.*

-	F	//
---	---	----

Replace the K2LCN or K4LCN board.*

-1	0	n
----	---	---

- n=(1 thru 7),
(-) or (C) Replace the K2LCN or K4LCN board; replace the module backplane or chassis assembly.
- n=(other) Replace the K2LCN or K4LCN board.*

-1	1	//
----	---	----

Replace the K2LCN or K4LCN board.*

-1	2	//
----	---	----

Replace the K2LCN or K4LCN board.*

-1	3	//
----	---	----

Replace the K2LCN or K4LCN board.*

-1	4	//
----	---	----

Replace the K2LCN or K4LCN board.*

-1	5	//
----	---	----

Replace the K2LCN or K4LCN board.*

-1	6	//
----	---	----

Replace the K2LCN or K4LCN board.*

-1	7	n
----	---	---

- n=(0), (1) Check for a duplicate node number; check the BNC connectors; Check/replace the LCN cable; replace the CLCNA (KLCNA) or CLCNB (KLCNB) board; replace the K2LCN or K4LCN board.
- n=(2 thru 5) Replace the K2LCN or K4LCN board; check/replace the K2LCN or K4LCN board or LCN board in another node.
- n=(6) Replace the Winchester (drive or interface).
- n=(7 thru 9) Reload the node; replace the K2LCN or K4LCN board.
- n=(other) Replace the K2LCN or K4LCN board or another board in the node that could be hanging up the bus.*

-1	8	n
----	---	---

- n=(0 thru 2) Replace the EPDG board.
- n=(3) reload the node; replace the K2LCN or K4LCN board.
- n=(4,5,8) Check for proper release version of the software; reboot the system.
- n=(9) Check/replace the LCN cable; check for proper release version of the software; replace the K2LCN or K4LCN board.
- n=(A,C,E,F) Suspect the load image - check the software release version and reload the node.
- n=(other) Replace the K2LCN or K4LCN board.*

-1	9	n
----	---	---

- n=(0,3,4,5) Reboot the system.
- n=(1 or 2) Record the error messages and call Honeywell TAC.
- n=(6) Use SMCC to find defective board.
- n=(7,8) Check memory capacity; replace the K2LCN or K4LCN board; check and reload the software.
- n=(9) Check and reload the software; replace the K2LCN or K4LCN board.
- n=A Replace the K2LCN or K4LCN board.
- n=(other) Replace the K2LCN or K4LCN board.*

-1	A	//
----	---	----

Replace the K2LCN or K4LCN board.*

-1	-	//
----	---	----

Replace the K2LCN or K4LCN board.*

-1	C	//
----	---	----

Replace the K2LCN or K4LCN board.*

-1		//
----	--	----

Replace the K2LCN or K4LCN board.*

-1	E	n
----	---	---

- n=(2) Replace the board in slot 2.
- n=(3) Replace the board in slot 3, if present.
- n=(other) Replace the K2LCN or K4LCN board.*

-1	F	n
----	---	---

- n=(0 thru 8),
(E) or (F) Replace the K2LCN or K4LCN board.
- n=(9) or (A) Replace the K2LCN or K4LCN board.*
- n=(-) or (C) Replace the CLCNA (KLCNA) or CLCNB (KLCNB) board; if the
error repeats, look for bus problems.
- n=(other) Check and/or replace the backplane or chassis assembly.

REPLACEMENT OF K2LCN WITH K4LCN BOARD

Appendix B

This appendix provides the procedure for replacing a K2LCN processor board with a K4LCN board in a Dual Node Module. It also includes the restrictions and conditions that apply to the replacement procedure.

B.1 INTRODUCTION

This procedure enables a trained technician to replace a 68020-based K2LCN processor board with a high density 68040-based K4LCN processor board in a Dual Node Module LCN node. The K4LCN board requires software release R500 or a later release. To be a candidate for this replacement, the module cannot contain a processor board that has special firmware (such as PCIM). If the module is one of a redundant pair, both processor boards must be replaced with K4LCNs boards. The K4LCN board provides a processor, LCN interface, clock translation, and contains four, eight or sixteen megawords of on-board memory.

WARNING

REPLACING A K2LCN BOARD WITH A K4LCN BOARD REQUIRES TURNING OFF POWER TO THE NODE, AND MUST BE PERFORMED AT AN APPROPRIATE TIME TO AVOID DISTURBANCE TO THE PROCESS AND POTENTIALLY HAZARDOUS CONSEQUENCES TO EQUIPMENT AND PERSONNEL. DETERMINE IN ADVANCE THAT THE SCHEDULING OF THIS PROCEDURE IS IN ACCORDANCE WITH PROPER SYSTEM OPERATION.

B.2 PREREQUISITE

The system must be up and running on Software Release 500 or a later release before performing the processor replacement.

B.3 NODE APPLICABILITY

Check that the K4LCN replacement board is applicable to the intended node.

For performance compatibility, processor board types in redundant node pairs should not be mixed. If you are replacing a K2LCN processor board and the affected node is one of a redundant pair, a K4LCN processor board must be installed in its partner also. Refer to subsection B.5.

B.4 MEMORY SIZE

Determine the amount of memory in the node, including all memory boards and any memory on the processor board that is being replaced. The replacement K4LCN board must have at least the same amount of memory. Having more memory is not a problem. Because the K4LCN board is available with different memory sizes, there are three upgrade kits available.

Upgrade Kit Description	Upgrade Kit Model Number	K4LCN Board Part Number	Memory Board Part Number
K4LCN Board – 4 Megawords	MP-ZPSC04	51402755-100	51201795-400
K4LCN Board – 8 Megawords	MP-ZPSC08	51402755-100	51201795-800
K4LCN Board – 16 Megawords	MP-ZPSC16	51402755-100	51201793-160

B.5 REDUNDANT MODULE REPLACEMENTS

This section applies only to a redundant Network Interface Module (NIM), Hiway Gateway (HG), or Enhanced Programmable Logic Controller Gateway (EPLCG).

Determine in which node, the primary or secondary, you want to perform the replacement first. The recommended procedure would have you replace the processor in the primary node first. This will require that you fail-over the primary to the secondary, making the primary available and allowing you to confirm good operation of both nodes. It is acceptable to perform the replacement to the secondary first, but you will not have validated the secondary's proper operation prior to the board changes. In either case, you **must be certain that the intended node is the backup**. When it is necessary to cause an intentional fail-over, perform the following:

At a Universal Station or Universal Station^X, perform a Shutdown of the intended node. This will cause an “orderly” shutdown to the “POWER ON” condition and notify all nodes that the unit is being shut down, as well as cause the backup to take over as the active node. If the backup, now the active primary, has successfully performed the “takeover,” you are ready to proceed to subsection B.6, step 2, below. When subsections B.6 through B. are completed on the first unit, repeat them on the second unit.

B.6 DISASSEMBLY

Use the following procedure to disassemble the Dual Node Module.

1. If the node is “ON LINE,” go to a Universal Station or Universal Station^X and execute a shutdown to the intended node. This will perform a “controlled” shutdown to the “POWER ON” condition and notify all nodes that the unit is being shut down.
2. Place the node’s ac power switch on the front of the module in the OFF position.
3. Remove the front cover by removing the screws and swinging the right side of the cover out 1 inch and then pulling it out to the right.
4. Be sure to observe the proper ESD prevention practices, especially while handling the K2LCN board.
5. Remove the K2LCN board from Slot 1 at the front of the module.

B.7 REASSEMBLY

Use the following procedure to reassemble the Dual Node Module.

1. Select the correct node address on the K4LCN board.

The LCN node address pinning block, TS2, is located at coordinate 21E on the early production K4LCN board assembly, 51401946-100. The LCN node address pinning block, TS1, is located at coordinate 7A on the later production K4LCN board assembly, 51402755-100. Refer to Figures 2-6 and 2-7, respectively, for pinning information.

2. Install the K4LCN board in Slot 1.
3. Fasten the replacement kit label at the rear of the module chassis next to the original label.
4. Replace the front cover.
5. Place the node’s ac power switch on the front of the module in the ON position.

B.8 TEST

Observe the indicators at the front of the module and especially on the K4LCN board. If a fault occurs, refer to Section 3 in this manual.

B.9 STARTUP

Restart the node, loading the node’s specific personality and database, as required. Refer to the *Engineer's Digest* manual.

Index

Topic	Section Heading
Address Pinning, K2LCN	2.8, Figure 2-4
Adjustment, Field	2.7
Alphanumeric Displays	A
Application Module Configuration	2.3, Tables 2-2, 2-3
Assembly	4.2
Background Shade Pinning, EPDGP I/O	2.8, Figure 2-5
Chassis Ground	2.6
Clock Pinning 2.9	
CLI Board, Troubleshooting	3.3.2
CNI Board, Troubleshooting	3.3.2
Computer Gateway Configuration	2.3, Tables 2-2, 2-3
Configurations, Basic Module	2.2, Table 2-1
Configurations, Specific Node	2.3, Tables 2-2, 2-3
DHI Board, Troubleshooting	3.3.2
Disassembly	4.1
Dual Node Module	1.1, Figure 1-1
Dual Node Module, Exploded View	4.1, Figure 4-1
EPDG Board, Troubleshooting	3.3.5
EPNI Board, Troubleshooting	3.3.6
EPNI2 Board, Troubleshooting	3.3.7
EPNI2 I/O Board Pinning	2.11
Fan Troubleshooting	3.3.1
FDC Board, Troubleshooting	3.3.2
Fiber Optic Extender Board (XLCNE2), Troubleshooting	3.3.7
Field Adjustment	2.7
Front Panel	2.4, Figure 2-3
General Description, Module	2.1
Ground, Logic and Chassis	2.6
Grounding, LCN Network	2.8
History Module Configuration	2.3, Tables 2-2, 2-3
Hiway Gateway Configuration	2.3, Table 2-3
Introduction	1
K2LCN/K4LCN Address Pinning	2.8, Figure 2-4
K2LCN/K4LCN Diagnostic Indicators	3.3, Figure 3-2
K2LCN/K4LCN Troubleshooting	3.3.4
K4KCN Replacement of K2LCN	B
LCN Network Grounding	2.8
Logic Ground 2.6	
Module Description	
General Description	2.1
Basic Module Configurations	2.2
Specific Node Configurations	2.3
Front Panel	2.4
Rear Panel Assembly	2.5
Logic and Chassis Ground	2.6
Field Adjustment	2.7
LCN Network Grounding	2.8
Network Interface Module Configuration	2.3, Tables 2-2, 2-3
NGI Board, Troubleshooting	3.3.9
NIM Board Pinning	2.10
Overview	1.1
Plant Network Module	2.3, Table 2-3
PLCI Board, Troubleshooting	3.3.3
PLC Gateway Configuration	2.3, Tables 2-2, 2-3

Index

Topic	Section Heading
Power On Test	3.2.1
Power Supply/Fan Troubleshooting	3.3.1
Power Supply Loading in XLCNE2 Host Node	2.11, Table 2-4
Power Supply Replacement	2.9
Precision Clock Pinning	2.9
Quality Logic Test	3.2.3
Rear Panel Assembly	2.5
Rear View, Two Dual Node Modules	2.2, Figure 2-2
Related Publications	1.2
Reset Test	3.2.2
SIO Board, Troubleshooting	3.3.2
Spare Parts	
Introduction	5.1
SPC Board, Troubleshooting	3.3.2
Startup	
Visual Checks	6.1
Initialize Module	6.2
TDC 3000 ^X Module Functions	2.1, Figure 2-1
Testing Flow Diagram	3.2, Figure 3-1
Test/Troubleshooting	3
Tests 3.1	
Test Procedures	
Power Up Test	3.2.1
Reset Test	3.2.2
Troubleshooting	
Power Supply/Fan	3.3.1
Controller Boards	3.3.2
PLCI Board	3.3.3
Processor Board (K2LCN)	3.3.4
EPDG Board	3.3.5
EPNI Board	3.3.6
Fiber Optic Extender Board (XLCNE2)	3.3.7
RS-485 to LCN Converter Boards (KLCNA, KLCNB)	3.3.8
NGI Board	3.3.9
Universal Station Configuration	2.3, Tables 2-2, 2-3
Universal Station with Floppy Drives Configuration	2.3, Table 2-3
Universal Station with Trend Pen Configuration	2.3, Table 2-3
XLCNE2 Troubleshooting	3.3.7

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